Electrical Machines

Module 3 Study Guide

Synchronous Machines

This guide combines all three source slide decks into one branded study PDF. It includes cleaned reading notes first and every original slide in the appendix so no material is lost.

Anthropic Brand Styling

Palette: `#141413`, `#FAF9F5`, `#B0AEA5`, `#E8E6DC`, `#D97757`, `#6A9BCC`, `#788C5D`. Headings use Poppins when available. Body text uses Lora when available, with serif fallback when not present on the system.

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Module Roadmap

This document is structured in two layers: a readable study section for revision, followed by a full appendix containing every original slide exactly as supplied.

Part 1: Generator Fundamentals

Principle of operation of the 3-phase synchronous generator, AC generation, EMF equation, and the main constructional classifications of alternators.

Part 2: Load Behaviour and Synchronisation

Armature reaction, synchronous impedance, phasor diagrams, voltage regulation by EMF and MMF methods, and synchronising conditions and lamp methods.

Part 3: Synchronous Motor

Construction, working principle, why the motor is not self-starting, starting methods, and the basic equivalent-circuit idea.

Study Notes

The note cards below are cleaned from the slide text for easier reading. Use the appendix whenever a page is mainly diagrammatic or whenever you want the exact original slide layout.

Part 1: Synchronous Generator Basics, EMF Equation and Construction

Principle of operation, AC generation, EMF equation, rotor and stator construction, and alternator classification.

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Part 1 • Slide 1 • Source: EM_Module 3_Part1 (1).pdf

This slide is primarily a title page or visual reference. The original slide is preserved in the appendix.

Three phase Synchronous Machine

Part 1 • Slide 2 • Source: EM_Module 3_Part1 (1).pdf

• Principle of Operation of 3-phase synchronous generator – classification -
• constructional features - EMF equation – coil-span factor and distribution factor (sinusoidal flux distribution only) – numerical problems Synchronous generator on no-load – open circuit characteristics –
• synchronous generator on load – armature reaction – effect of armature reaction
• Equivalent circuit - phasor diagram – voltage regulation –
• predetermination of voltage regulation by emf and mmf method –
• Parallel operation - synchronous generator on infinite bus-bar – conditions – methods of synchronisation – dark lamp method – bright lamp method (2 hours)
• Synchronous motor – rotating magnetic field (no derivation) - principle of operation – starting methods

• Principle of Operation of 3-phase synchronous generator

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AC Generator

Part 1 • Slide 4 • Source: EM_Module 3_Part1 (1).pdf

• Faraday's law dictates that if a coil of N turns experiences a change in magnetic flux, then the induced voltage V is given by $d\Phi$ $V = N$ dt
• If a coil of area A rotates with respect to a field B, and if at a particular time it is at an angle θ to the field, then the flux linking the coil is $BA\cos\theta$, and the rate of change of flux is given by $d\Phi$ $d(\sin\theta)$ $d\theta = BA = \cos\theta = \omega\cos\theta$ dt dt dt

Magnetic induction in a wire moving in a field.

Part 1 • Slide 5 • Source: EM_Module 3_Part1 (1).pdf

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Fleming's Right hand rule

Part 1 • Slide 6 • Source: EM_Module 3_Part1 (1).pdf

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Fleming’s Right hand rule Used to determine the direction of emf induced

Part 1 • Slide 7 • Source: EM_Module 3_Part1 (1).pdf

in a conductor The middle finger , the fore finger and thumb of
the left hand are kept at right angles to one
another. The fore finger represent the direction of magnetic field The thumb represent the direction of motion of the conductor The middle finger will indicate the direction of the inducted emf . This rule is used in DC Generators

AC generator with slip rings and brushes.

Part 1 • Slide 8 • Source: EM_Module 3_Part1 (1).pdf

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• Thus for the arrangement shown below dPhi d(sintheta) V =N = NBA = NBAomega costheta dt dt dPhi V =N dt

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• Wires connected to the rotating coil would get twisted

Part 1 • Slide 10 • Source: EM_Module 3_Part1 (1).pdf

• Therefore we use circular slip rings with sliding contacts called brushes

• Therefore this arrangement produces a sinusoidal output as shown below

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SYNCHRONOUS GENERATOR

Part 1 • Slide 12 • Source: EM_Module 3_Part1 (1).pdf

CONSTRUCTION
• rotor carry the “field windings” , while “armature windings” (or “stator windings”) carry the main voltages of machine in synchronous generator, rotor winding energized by dc source to develop rotor magnetic field
- rotor is turned by a prime mover, producing a rotating magnetic field which induce 3 phase voltages in stator windings

therefore:

- rotor windings - field windings
- stator windings - armature windings

3-phase synchronous generator – classification

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Based on Prime Mover • Hydro Generators – Driven by water turbines • Turbo Generators – Driven by steam turbines • Diesel Engine Generators – Driven by diesel engines • Gas Turbine Generators – Driven by gas turbines

Part 1 • Slide 14 • Source: EM_Module 3_Part1 (1).pdf

Based on No. of Phase • Single phase •Three phase
Based on type of roatating • Rotating Armature •Rotating Field
Based on Rotor Construction •Salient Pole machine •Non Salient Pole Machine

Advantages of rotating field and stationary armature type

Part 1 • Slide 15 • Source: EM_Module 3_Part1 (1).pdf

High voltage generation
Better insulation
Rotor weight – less
Current collection - easy
Lesser number of slip rings

Alternator - Construction

Part 1 • Slide 16 • Source: EM_Module 3_Part1 (1).pdf

Standard construction consists of : Armature windings on stationery stator Field poles on rotor. The field windings are excited by DC supply from the exciter and given through the slip rings

Types of alternators based on rotor construction

Part 1 • Slide 17 • Source: EM_Module 3_Part1 (1).pdf

Salient pole alternator
Non salient pole or Turbo or
Cylindrical rotor alternator

Rotor Non salient pole

Part 1 • Slide 18 • Source: EM_Module 3_Part1 (1).pdf

Salient pole alternator or Turbo or cylindrical rotor alternator

• Rotor of synchronous machine can be

Part 1 • Slide 19 • Source: EM_Module 3_Part1 (1).pdf

Nonsalient: 2 pole rotor Salient: six-pole rotor

Salient pole type alternator

Part 1 • Slide 20 • Source: EM_Module 3_Part1 (1).pdf

It consists of cast iron frame which supports the
silicon steel laminated (bundling of thin sheets)
armature core .
The core has slots on its inner periphery for housing
the conductor .
Silicon steel material reduces the hysteresis loss
and the lamination structure reduces the eddy current
loss.
Suitable for low and medium speed alternator
(Ex Hydro alternator)

Advantages of a smooth cylindrical type rotor

Part 1 • Slide 21 • Source: EM_Module 3_Part1 (1).pdf

Less windage loss
No need of damper winding
Good balancing of the rotor
Sound less running
Good cooling to the field because it is a
distributed one.

SYNCHRONOUS GENERATOR

Part 1 • Slide 22 • Source: EM_Module 3_Part1 (1).pdf

CONSTRUCTION
• Photograph of a salient 8-pole synchronous machine rotor

SYNCHRONOUS GENERATOR

Part 1 • Slide 23 • Source: EM_Module 3_Part1 (1).pdf

CONSTRUCTION
• Rotor experience varying magnetic fields, therefore is constructed of thin laminations to reduce eddy current losses
• To supply the rotor winding while it is rotating, special arrangement employed to connect its terminal to dc supply
1. supply dc power from an external dc source to rotor by means of slip rings
2. supply dc power from a special dc power source mounted on shaft of rotor

SYNCHRONOUS GENERATOR

Part 1 • Slide 24 • Source: EM_Module 3_Part1 (1).pdf

CONSTRUCTION
• Rotor of synchronous machine can be
Nonsalient: 2 pole rotor Salient: six-pole rotor

Non-Salient Pole (Cylindrical Salient Pole Rotor Rotor)

Part 1 • Slide 25 • Source: EM_Module 3_Part1 (1).pdf

Rotor Construction Projecting poles mounted on rotor Smooth solid cylindrical rotor
Air Gap Non-uniform air gap Uniform air gap Small diameter, long axial
Shape Large diameter, short axial length length
Speed of Operation Low speed (100–600 rpm) High speed (1500–3000 rpm) Large number of poles (6–60
Number of Poles Usually 2 or 4 poles poles)
Mechanical Strength Lower mechanical strength Very high mechanical strength Thermal & nuclear power
Applications Hydroelectric power plants plants Hydrogen or forced cooling (in

Cooling Natural or air cooling turbo generators)

SYNCHRONOUS GENERATOR

Part 1 • Slide 26 • Source: EM_Module 3_Part1 (1).pdf

CONSTRUCTION
• salient 8-pole synchronous machine rotor

SYNCHRONOUS GENERATOR

Part 1 • Slide 27 • Source: EM_Module 3_Part1 (1).pdf

CONSTRUCTION
• Rotor experience varying magnetic fields, therefore is constructed of thin laminations to reduce eddy current losses
• To supply the rotor winding while it is rotating, special arrangement employed to connect its terminal to dc supply
1. supply dc power from an external dc source to rotor by means of slip rings
2. supply dc power from a special dc power source mounted on shaft of rotor

Slide 28

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SYNCHRONOUS GENERATOR Speed of rotation of synchronous generator

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• synchronous generators are synchronous, during their operation
means: electrical frequency is synchronized with mechanical speed of rotor
• Relation between electrical frequency of stator and mechanical speed of rotor as shown before:
• f =Ns p / 120 f : electrical frequency in Hz Ns: speed of rotor in r/min p: number of poles

Winding factor

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Pitch factor

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Distribution factor

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Slide 33

Part 1 • Slide 33 • Source: EM_Module 3_Part1 (1).pdf

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EMF Equation

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Slide 35

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Part 2: Load Operation, Regulation and Synchronisation

Armature reaction, synchronous impedance, phasor diagrams, voltage regulation, and methods of synchronisation.

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Part 2 • Slide 1 • Source: EM_Module 3_Part2.pdf

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Three phase Synchronous Machine

Part 2 • Slide 2 • Source: EM_Module 3_Part2.pdf

- Synchronous generator on no-load – open circuit characteristics –
- synchronous generator on load – armature reaction – effect of armature reaction
- Equivalent circuit - phasor diagram – voltage regulation –
- predetermination of voltage regulation by emf and mmf method
- Parallel operation - synchronous generator on infinite bus-bar –
- conditions – methods of synchronisation – dark lamp method – bright lamp method

Leakage reactance

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Armature Reaction

Part 2 • Slide 4 • Source: EM_Module 3_Part2.pdf

- Effect of armature flux on main flux
- In alternator it depend on type of load
- Effects are • Magnetising • Demagnetising • Cross magnetising

i) At upf load

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ii) Zero power factor lagging

Part 2 • Slide 6 • Source: EM_Module 3_Part2.pdf

This slide is primarily a title page or visual reference. The original slide is preserved in the appendix.

iii) Zero power factor leading

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Synchronous Impedance

Part 2 • Slide 8 • Source: EM_Module 3_Part2.pdf

- Synchronous Reactance
- Synchronous Impedance

Alternator on load Phasor Diagram

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Equivalent Circuit

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Slide 11

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Load Characteristics

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Voltage Regulation

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EMF Method (Synchronous Impedance Method)

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Slide 15

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MMF (Ampere Turn Method)

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Parallel Operation of Alternators

Part 2 • Slide 17 • Source: EM_Module 3_Part2.pdf

Efficiency
Reliability or Continuity of Service
Maintenance and Repair
Physical Size

Requirements for Parallel Operation

Part 2 • Slide 18 • Source: EM_Module 3_Part2.pdf

• Alternators must have same output voltage rating
• Speeds of the machines should be such as to generate same frequency
• Alternators should be of same type so as to generate voltages of same waveform
• Prime movers of the alternators should have same speed-load characteristics
• Alternators should have reactances in their armature

Conditions for Synchronising

Part 2 • Slide 19 • Source: EM_Module 3_Part2.pdf

- Synchronising - The process of connecting an alternator in parallel with another alternator or with the common bus bar.
- For satisfactory operation,

• The terminal voltage of incoming machine must be exactly equal to that of running machines or common bus bar

• The speed of incoming machine must be such that its frequency f equals to bus bar frequency
• Phase of incoming machine voltage must be the same as that of the bus bar voltage relative to the load.
• For three phase alternator, the phase sequence of incoming machine must be same as that of bus bar.

Methods of Synchronisation

Part 2 • Slide 20 • Source: EM_Module 3_Part2.pdf

Synchronising of three phase alternators 1 • Three Dark Lamp Method 2 • Bright lamp Method (Two Bright and One Dark Lamp Method) • Synchroscope

Lamp Method Synchronisation of Single-Phase Alternators Figure 8 : Synchronising of single phase alternators

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Dark Lamp Method

Part 2 • Slide 22 • Source: EM_Module 3_Part2.pdf

Equality of terminal voltages can be determined by connecting voltmeters to the incoming and running alternators.
Equality of frequency and phasing can be determined with the help of synchronising lamps.
Let alternator B is to be paralleled with alternator A,
Prime mover of Alternator B is started and brought up close to the rated speed.
Alternator B is excited and its voltage is raised by increasing the excitation.
If frequencies of alternators A & B are exactly same and their terminal voltages w.r.t local series circuit -> no resultant voltage will act across lamps L1 and L2 -> Lamps will remain dark.
If frequencies of A & B are not equal -> flickering of lamps. Frequency of flickering = Difference of frequencies of two alternators.
==> greater the difference in frequencies, greater will be the frequency of flicker

Dark Lamp Method

Part 2 • Slide 23 • Source: EM_Module 3_Part2.pdf

Synchronising is done at the middle of dark period.
”At the time of synchronising of alternators, the speed of the incoming machine is adjusted until the lamps go in and out very slowly, the terminal voltage of the incoming alternator is made equal to bus bar voltage by adjusting the excitation of the incoming alternator and then switch of incoming alternator is closed in the middle of the dark period.”
But it is not so easy to judge the middle of dark period.

Bright Lamp Method Figure 9 : Synchronising of single phase alternators

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Bright Lamp Method

Part 2 • Slide 25 • Source: EM_Module 3_Part2.pdf

Easy to judge the middle of bright period
In Bright lamp method, lamps are cross-connected.
Maximum voltage across lamps will occur when alternator C is in phase opposition with alternator A.
Magnitude of maximum voltage that can exist across a lamp is double that of the voltage of one machine.
After doing the adjustments, the synchronising switch is closed at the middle of bright period.

Synchronization of 3-Phase Alternators

Part 2 • Slide 26 • Source: EM_Module 3_Part2.pdf

”Phasing out the alternator” -> The process of checking the phase sequence and getting it correct
1 Three Dark Lamp Method
2 Two Bright and One Dark Lamp Method
3 Using Synchroscope

Three Dark Lamp Method Figure 10 : Synchronising of 3-Phase Alternators

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Three Dark Lamp Method

Part 2 • Slide 28 • Source: EM_Module 3_Part2.pdf

Prime mover of alternator B is started and brought up close to the rated speed.
Then alternator B is excited and its voltage is raised by increasing the excitation.
If the incoming alternator B is properly connected, all the three lamps should become bright and dark together.
If they bright and dim in sequence ==> incoming alternator B is not properly connected with the bus-bars and the phase sequence of incoming alternator B must be reversed relative to the system.(Interchange any two leads on either the alternator side or the line side of the switch)
The speed of incoming machine is further adjusted until the lamps flicker at a very rate and voltage is made equal to the bus bar voltage by adjusting the excitation.
The synchronising switch is closed at the instant all the three lamps

Two Bright and One Dark Lamp Method Figure 11 : Synchronising of 3-Phase Alternators

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Two Bright and One Dark Lamp Meth When incoming alternator is in synchronism with running machine, lamps L1 and L3 are bright and L2 is dark. Since near the point of synchronism, the brightness of one lamp increases and of another decreases. The instant at which the incoming machine is in synchronism with the bus-bars can be accurately determined and the paralleling switch will be closed at this instant. If the incoming machine is too fast to voltage across L3 decreases, L1 increases and L2 decreases. If the incoming machine is too slow to voltage across L3 increases, L1 decreases and L2 increases. Hence when the three lamps are placed in a ring, a light wave travelling in counter-clockwise direction indicates that the incoming machine is slow and light wave travelling in clockwise direction indicates that the incoming machine is fast. Synchronising switch is closed when changes in light are very slow and at the instant the lamp L is2dark. Dept. of EEE, MACE) Synchronous and Induction Machines May 1

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Slide 31

Part 2 • Slide 31 • Source: EM_Module 3_Part2.pdf

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Using Three Limbed Transformer and Lamp Figure 12 : Three Limbed Transformer and Lamp

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Synchroscope

Part 2 • Slide 33 • Source: EM_Module 3_Part2.pdf

Synchroscope -> Instrument indicating the difference of phase and frequency between two voltages.
Its a split-phase motor
Torque is developed if the frequencies of the two voltages differ.
Voltages from corresponding phases of incoming and running alternators are applie to synchroscope.
When frequencies are equal, no torque is exerted on the pointer and the pointer stops at the middle of the dial.
When pointerstops at vertical position, the frequencies are equal, the voltages are in phase and the paralleling switch can be closed.

Synchronising Current Figure 15 : Voltage Phasors

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Synchronising Current

Part 2 • Slide 35 • Source: EM_Module 3_Part2.pdf

When two alternators are in synchronism -> have equal emfs in exact phase opposition -> no circulating current (Fig. (a))
When induced emfs are equal in magnitude but not in exact phase opposition -> resultant emf around the local circuit /= zero -> causes flow of current -> Synchronising current(Isy)
-> Let alternator-2 tends to retard,
E2 falls back by a phase angle δ electrical degrees
|E1| = |E2| = E
but phase difference = 180 - δ

to Resultant emf(ER) 180 - ER = 2ECos delta 2 delta = 2ECos 90 - 2 = 2ESindelta delta = 2E × = Edelta to Synchroni sing ER Edelta current(I I s = sy = y) Zcs Zcs

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Synchronising Current

Part 2 • Slide 37 • Source: EM_Module 3_Part2.pdf

Isy lags behind ER by an angle θ
If Rce «< Xcs ,
ER Isy = Xcs

==> Isy lags ER by 90o ==> almost in phase with E1
Isy -> generating current w.r.t machine 1 and motoring current w.r.t machine 2
Isy sets up synchronising torque(Tsy) which tends to accelerate machine no.2 and decelerate machine no.1
Any departure from synchronism results in development of synchronising torque which tends to keep the machines in synchronism

Synchronising Power

Part 2 • Slide 38 • Source: EM_Module 3_Part2.pdf

Power supplied by Machine no.1 = E1Isy CosΦ1
Power received by Machine no.2 = E2Isy Cos (180 − Φ2)
Power supplied by Machine no.1 = Power received by Machine no.2 + copper losses
Power supplied by Machine no.1 is called as Synchronising power(Psy) Psy = E1Isy CosΦ1 = E1Isy Eδ =E× X cs δE 2 = Xcs
Total synchronising power = 3P sy = 3δE Xcs

Synchronising Torque 2πNs 3Psy = T sy ×

Part 2 • Slide 39 • Source: EM_Module 3_Part2.pdf

Therefore Synchronising torque, 3Psy × 60 sy T = 2πNs
where Ns is the synchronous speed in rpm

Effects of Changing Excitation Figure 16 : (a) Figure 17 : (b)

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Part 3: Synchronous Motor Construction and Starting

Construction of the synchronous motor, principle of operation, failure to self-start, starting methods, and equivalent circuit.

MODULE III

Part 3 • Slide 1 • Source: EM_Module 3_Part3.pdf

Synchronous motor – Construction, Principle of operation, Effect of excitation on armature

Slide 2

Part 3 • Slide 2 • Source: EM_Module 3_Part3.pdf

This slide is primarily a title page or visual reference. The original slide is preserved in the appendix.

Construction of Synchronous motor

Part 3 • Slide 3 • Source: EM_Module 3_Part3.pdf

- It has mainly 2 parts

-

1. Stator
2. Rotor

1. Stator or Armature

Part 3 • Slide 4 • Source: EM_Module 3_Part3.pdf

- Stator is the stationary part
- Main parts of stator are stator core, stator winding & stator frame
- Stator core contain an iron ring made of silicon steel laminations
- Slots are cut on the inner periphery of the stator core, in which stator conductors are placed
- Stator core is laminated to reduce eddy current loss
- axial & radial ventilating ducts to provide efficient cooling
- Slots can be open, semi closed or totally closed
- Open slots are commonly used

Stator contain a 3 phase winding which is normally star connected

Part 3 • Slide 5 • Source: EM_Module 3_Part3.pdf

- Stator conductors are made of copper or aluminium
- Windings can be single layer or double layer, lap or wave, concentrated or distributed
- The whole structure is held in a frame made of cast iron or welded steel plate

2. Rotor or Field magnet system

Part 3 • Slide 6 • Source: EM_Module 3_Part3.pdf

- It is the rotating part of the machine
- Rotor contain Field magnets & are supplied with DC voltage
- For this external DC sources are used & exciting current is supplied to rotor through slip rings & brushes
- An exciter mounted on the motor shaft can also be used to provide necessary excitation.
- Normally, the rotor is having a salient pole structure
- Field poles contains laminated pole core & pole shoe (to reduce eddy current losses) and field windings

Slide 7

Part 3 • Slide 7 • Source: EM_Module 3_Part3.pdf

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Now stator North pole (Ns) & rotor North pole (Nr) will repel each other & rotor experiences a torque in anticlockwise direction

Part 3 • Slide 8 • Source: EM_Module 3_Part3.pdf

- After a period of half cycle, stator poles changes their position but rotor poles remain at previous position
- Now stator South pole (Ss) & rotor North pole (Nr) will attract each other & rotor experiences a torque in clockwise direction
- Since stator poles changes their polarity rapidly, rotor experiences torque in opposite directions in a cycle.
- Due to high inertia of rotor, motor fails to start

How to get unidirectional torque?

Part 3 • Slide 9 • Source: EM_Module 3_Part3.pdf

- If rotor is rotated by some external means at a speed near synchronous speed, in the same direction of stator magnetic field, then rotor will experience a unidirectional torque
- Now stator South pole (Ss) & rotor North pole (Nr) will attract each other & rotor experiences a torque in clockwise direction
- After a period of half cycle, stator & rotor magnetic field poles changes their position as shown in second figure
- rotor experiences a torque in clockwise direction.

Under this condition, stator & rotor poles of opposite polarity experiences a strong attraction & the mutual attraction locks the rotor & stator together

Part 3 • Slide 10 • Source: EM_Module 3_Part3.pdf

- Now if the external prime mover is removed, rotor will continue to rotate at synchronous speed due to magnetic locking between stator & rotor

Starting methods of Synchronous motor

Part 3 • Slide 11 • Source: EM_Module 3_Part3.pdf

- Synchronous motor has no self starting torque & motor must first be brought upto synchronous speed by some external means for getting unidirectional torque in rotor.
- Various methods of starting of synchronous motors are

1. From DC source

- If a DC supply & a DC motor (shunt/compound) is available, the Synchronous motor is coupled to DC motor & is started using DC motor
- DC motor is coupled to Synchronous motor and dc motor is started & rotated at synchronous speed
- AC supply is given to Synchronous motor (make sure that rotating magnetic field & rotor of synchronous motor are rotating in same direction)
- Now DC field excitation is given & rotor gets locked to stator rotating magnetic field & continue to rotate at synchronous speed
- The DC motor is now disconnected from supply

2. By using an AC motor

Part 3 • Slide 12 • Source: EM_Module 3_Part3.pdf

- Here a small AC induction motor (called pony motor) is used for starting the Synchronous motor
- First the induction motor is started & will rotate Synchronous motor rotor at near synchronous speed
- AC supply is given to Synchronous motor (make sure that rotating magnetic field & rotor of synchronous motor are rotating in same direction)
- Now DC field excitation is given & rotor gets locked to stator rotating magnetic field & continue to rotate at synchronous speed
- The Induction motor is now disconnected from supply

3. By providing damper windings

- Synchronous motor can be made self starting by providing a special winding on rotor poles known as damper wind-

ing or squirrel cage winding
- Damper winding consist of short circuited copper bars placed in rotor poles
- When AC supply is given to stator, the synchronous motor will start as an Induction motor & rotates at a speed near synchronous speed
- Now DC field excitation is given & rotor gets locked to stator rotating magnetic field & continue to rotate at synchronous speed

Equivalent circuit of a Synchronous motor

Part 3 • Slide 13 • Source: EM_Module 3_Part3.pdf

- In a synchronous motor we are giving two supply, DC supply to rotor & 3 phase AC supply to stator
- When AC supply is given to stator, a current flows through stator windings
- Stator windings have resistance (Ra) & synchronous reactance (Xs)
- Also when a DC supply is given to rotor, a magnetic field is produced & it links stator conductors,
- As rotor rotates, flux linked with stator conductors changes & an EMF induces in stator windings. This is called back EMF (Eb) or counter EMF
- The back EMF opposes the stator supply voltage

Original Slide Appendix

Every source slide is embedded below for exact reference. This appendix preserves diagrams, formula layouts, and any visual content that may not extract cleanly as text.

Part 1: Original Slides

Source: EM_Module 3_Part1 (1).pdf

PCEOT402
ELECTRICAL MACHINES

Module 3

Synchronous Machines

Source: EM_Module 3_Part1 (1).pdf

Three phase Synchronous Machine

- Principle of Operation of 3-phase synchronous generator – classification -
- constructional features - EMF equation – coil-span factor and distribution factor (sinusoidal flux distribution only) – numerical problems Synchronous generator on no-load – open circuit characteristics –
- synchronous generator on load – armature reaction – effect of armature reaction
- Equivalent circuit - phasor diagram – voltage regulation –
- predetermination of voltage regulation by emf and mmfmethod –
- Parallel operation - synchronous generator on infinite bus-bar – conditions – methods of synchronisation – dark lamp method – bright lamp method (2 hours)
- Synchronous motor – rotating magnetic field (no derivation) - principle of operation – starting methods

Part 1 • Slide 3

Source: EM_Module 3_Part1 (1).pdf

- Principle of Operation of 3-phase synchronous generator

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AC Generator

- Faraday's law dictates that if a coil of *N* turns experiences a change in magnetic flux, then the induced voltage *V* is given by

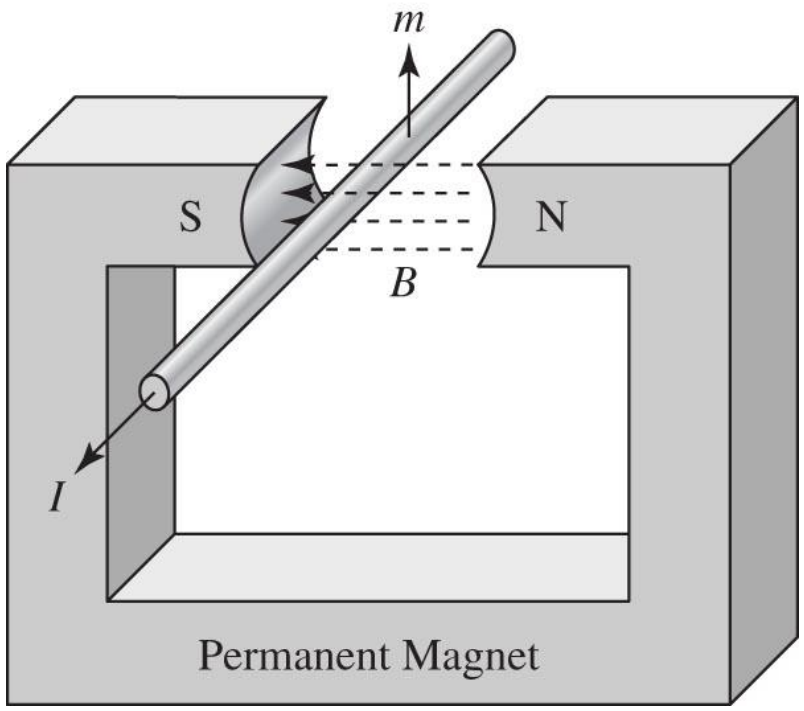
$$V = N\frac{\mathrm{d}\Phi}{\mathrm{d}t}$$

- If a coil of area *A* rotates with respect to a field *B*, and if at a particular time it is at an angle θ to the field, then the flux linking the coil is $BA\cos\theta$, and the rate of change of flux is given by

$$\frac{d\Phi}{dt} = BA\frac{\mathrm{d}(\sin\theta)}{\mathrm{d}t} = \frac{\mathrm{d}\theta}{\mathrm{d}t}\cos\theta = \omega\cos\theta$$

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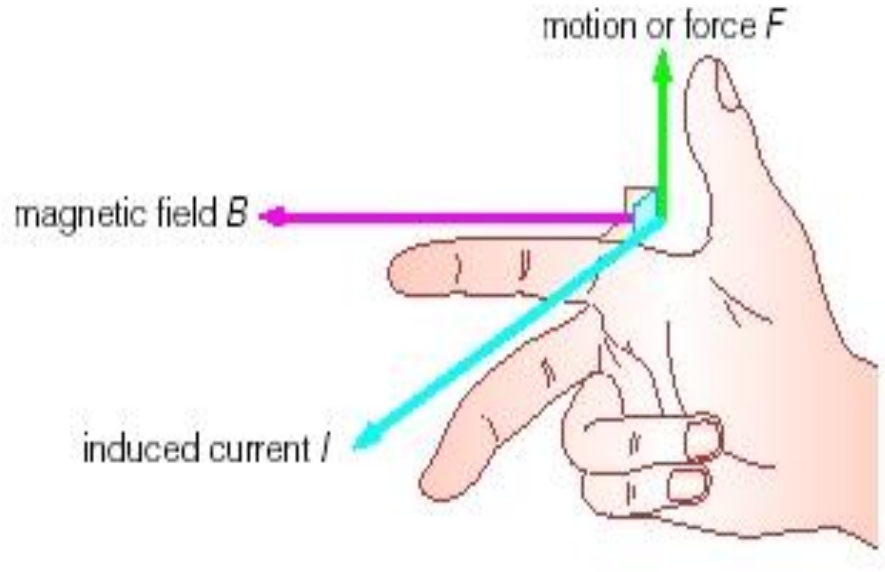
Magnetic induction in a wire moving in a field.



5

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Fleming’s Right hand rule



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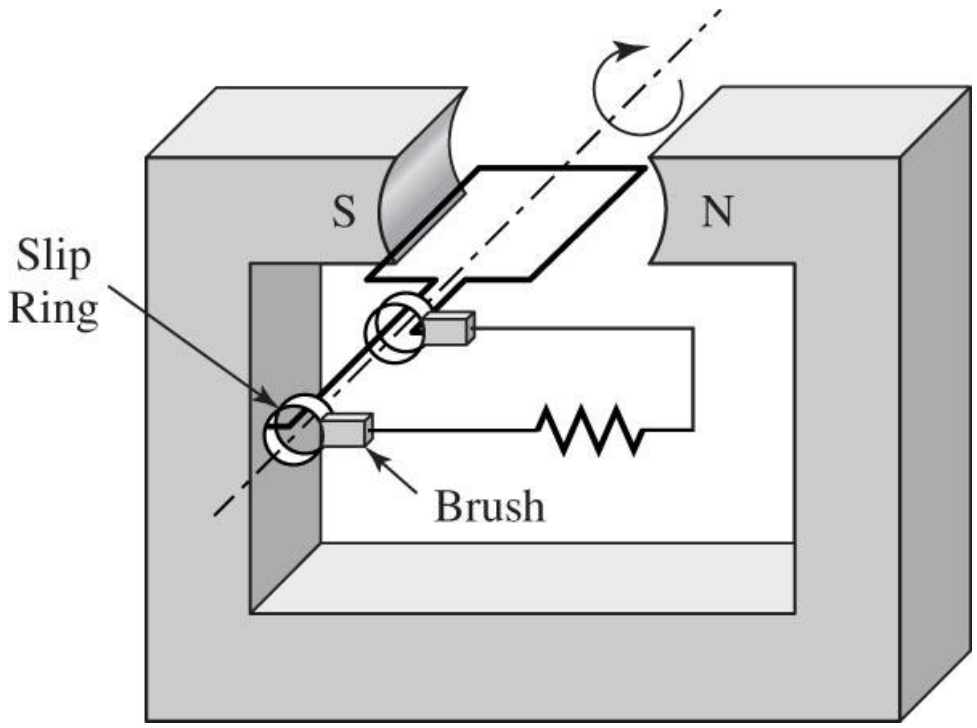
Fleming's Right hand rule

- Used to determine the direction of emf induced in a conductor
- The middle finger , the fore finger and thumb of the left hand are kept at right angles to one another.
  - The fore finger represent the direction of magnetic field
  - The thumb represent the direction of motion of the conductor
  - The middle finger will indicate the direction of the inducted emf .

This rule is used in DC Generators

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AC generator with slip rings and brushes.

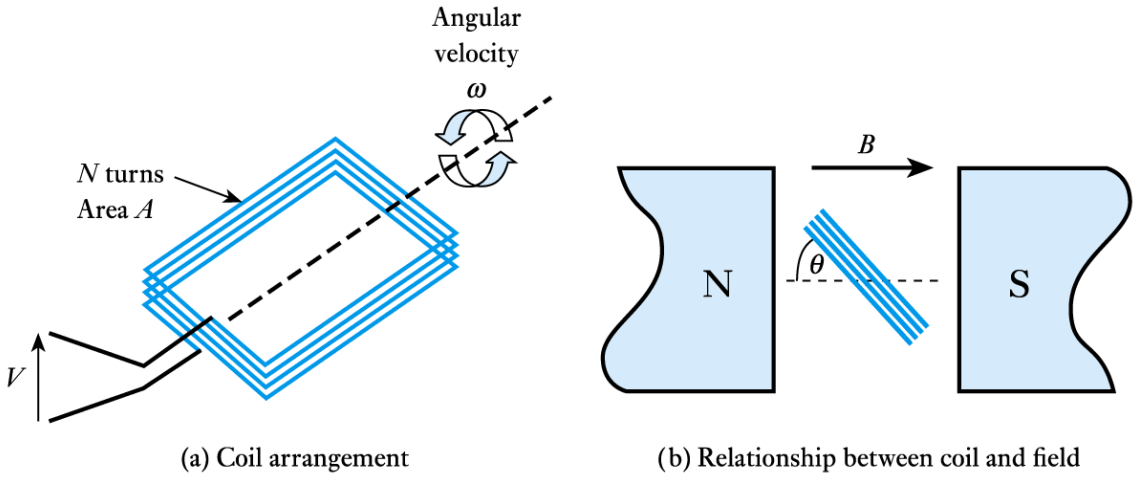


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- Thus for the arrangement shown below

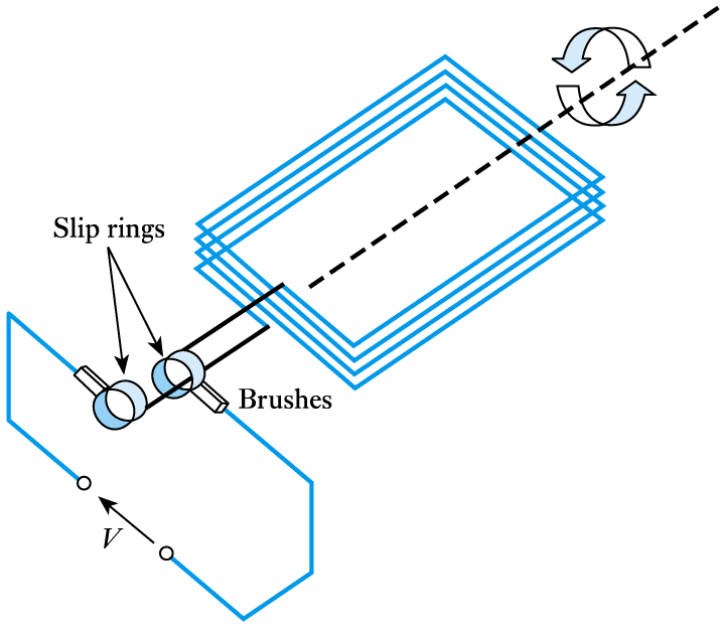
$$V = N\frac{d\Phi}{dt} = NBA\frac{d(\sin\theta)}{dt} = NBA\omega\cos\theta$$



(a) Coil arrangement (b) Relationship between coil and field

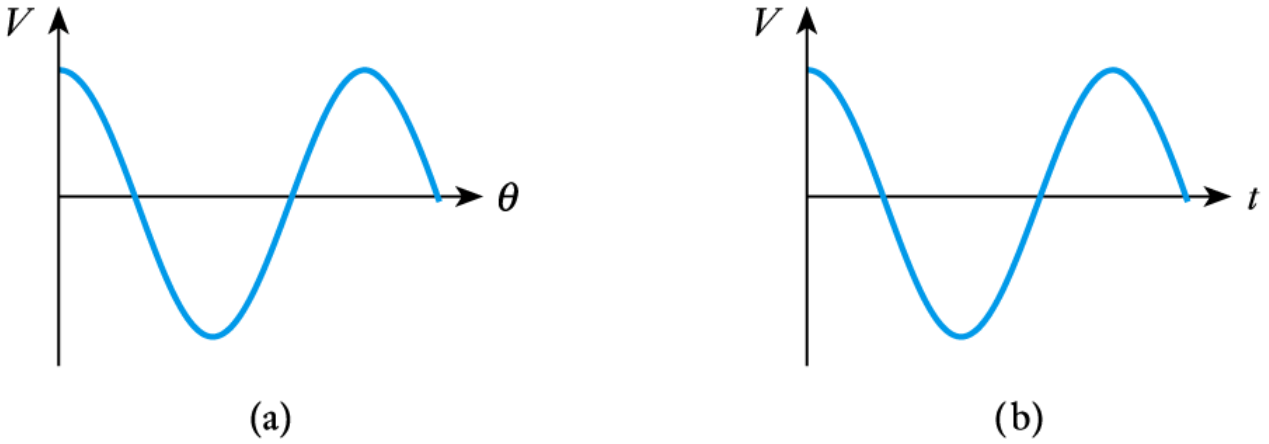
Source: EM_Module 3_Part1 (1).pdf

- Wires connected to the rotating coil would get twisted
- Therefore we use circular **slip rings** with sliding contacts called **brushes**



Source: EM_Module 3_Part1 (1).pdf

- Therefore this arrangement produces a sinusoidal output as shown below



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SYNCHRONOUS GENERATOR CONSTRUCTION

- **rotor carry the "field windings" , while "armature windings" (or "stator windings") carry the main voltages of machine**

  in synchronous generator, rotor winding energized by dc source to develop rotor magnetic field

- rotor is turned by a prime mover, producing a rotating magnetic field which induce 3 phase voltages in stator windings

therefore:

- rotor windings - field windings
- stator windings - armature windings

Source: EM_Module 3_Part1 (1).pdf

3-phase synchronous generator – classification

Source: EM_Module 3_Part1 (1).pdf

Based on Prime Mover

- **Hydro Generators** – Driven by water turbines
- **Turbo Generators** – Driven by steam turbines
- **Diesel Engine Generators** – Driven by diesel engines
- **Gas Turbine Generators** – Driven by gas turbines

Based on No. of Phase

- Single phase
- Three phase

Based on type of roatating

- Rotating Armature
- Rotating Field

Based on Rotor Construction

- Salient Pole machine
- Non Salient Pole Machine

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Advantages of rotating field and stationary armature type

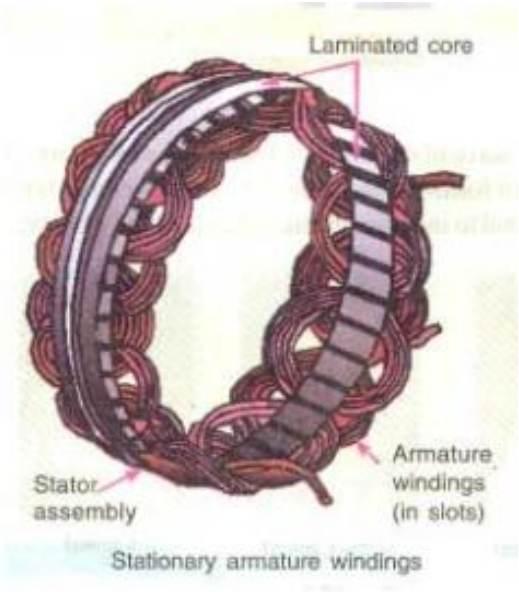
- High voltage generation
- Better insulation
- Rotor weight – less
- Current collection - easy
- Lesser number of slip rings

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Alternator - Construction

Standard construction consists of :

- Armature windings on stationery stator
- Field poles on rotor.
- The field windings are excited by DC supply from the exciter and given through the slip rings



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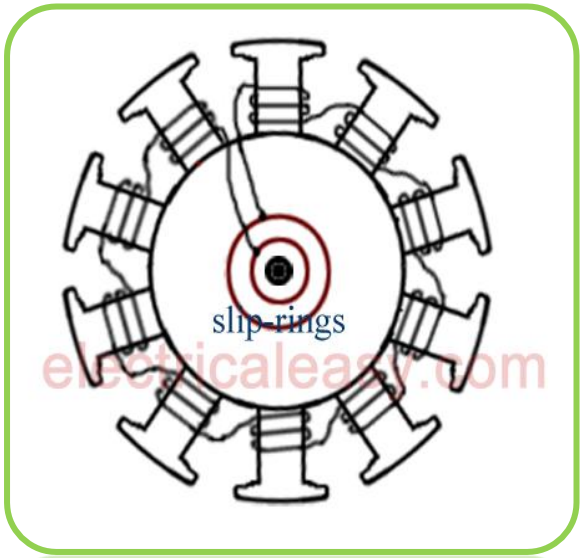
Types of alternators based on rotor construction

- Salient pole alternator
- Non salient pole or Turbo or Cylindrical rotor alternator

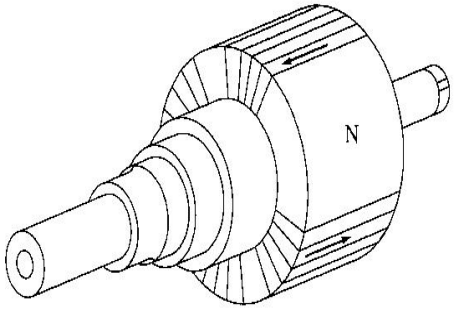
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Rotor

Salient pole alternator



Non salient pole or Turbo or cylindrical rotor alternator

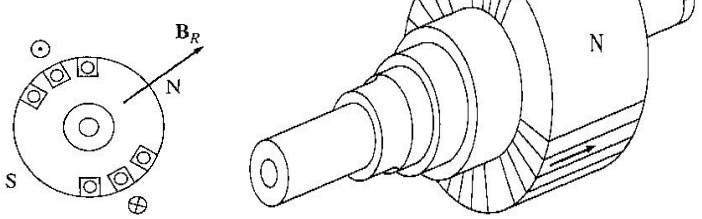


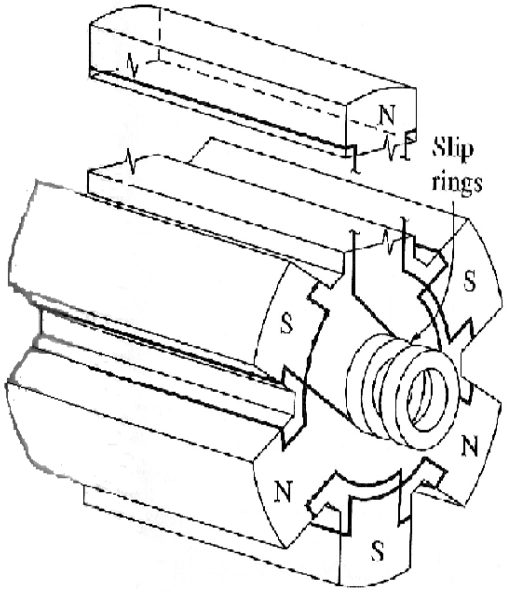
Side View

Source: EM_Module 3_Part1 (1).pdf

- Rotor of synchronous machine can be

Nonsalient: 2 pole rotor **Salient:** six-pole rotor





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Salient pole type alternator

- It consists of cast iron frame which supports the silicon steel laminated (bundling of thin sheets) armature core .
- The core has slots on its inner periphery for housing the conductor .
- Silicon steel material reduces the hysteresis loss and the lamination structure reduces the eddy current loss.
- Suitable for low and medium speed alternator (Ex Hydro alternator)

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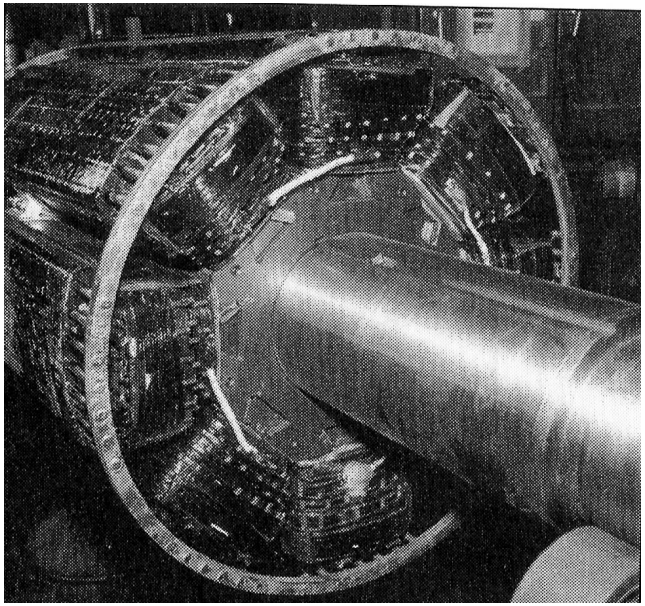
Advantages of a smooth cylindrical type rotor

- Less windage loss
- No need of damper winding
- Good balancing of the rotor
- Sound less running
- Good cooling to the field because it is a distributed one.

Source: EM_Module 3_Part1 (1).pdf

SYNCHRONOUS GENERATOR CONSTRUCTION

- Photograph of a salient 8-pole synchronous machine rotor



Source: EM_Module 3_Part1 (1).pdf

SYNCHRONOUS GENERATOR CONSTRUCTION

- **Rotor** experience varying magnetic fields, therefore is constructed of thin laminations to reduce eddy current losses
- To supply the rotor winding while it is rotating, special arrangement employed to connect its terminal to dc supply

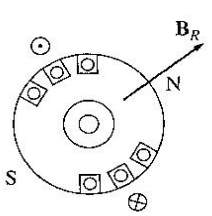
1. supply dc power from an external dc source to rotor by means of slip rings
2. supply dc power from a special dc power source mounted on shaft of rotor

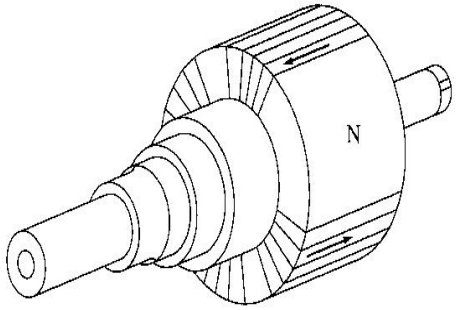
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SYNCHRONOUS GENERATOR CONSTRUCTION

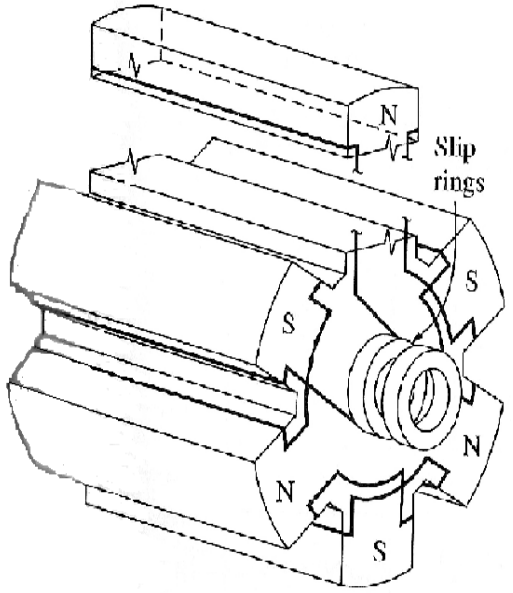
- Rotor of synchronous machine can be

Nonsalient: 2 pole rotor **Salient:** six-pole rotor





Side View



Part 1 • Slide 25

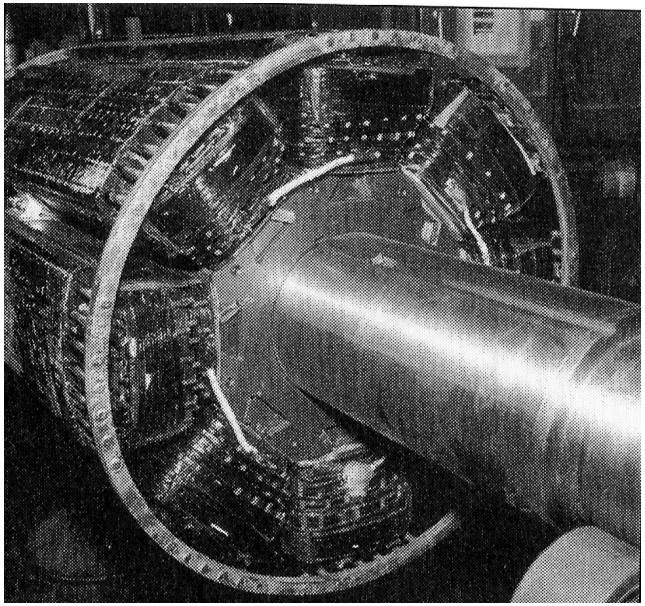
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	Salient Pole Rotor	Non-Salient Pole (Cylindrical Rotor)
Rotor Construction	Projecting poles mounted on rotor	Smooth solid cylindrical rotor
Air Gap	Non-uniform air gap	Uniform air gap
Shape	Large diameter, short axial length	Small diameter, long axial length
Speed of Operation	Low speed (100–600 rpm)	High speed (1500–3000 rpm)
Number of Poles	Large number of poles (6–60 poles)	Usually 2 or 4 poles
Mechanical Strength	Lower mechanical strength	Very high mechanical strength
Applications	Hydroelectric power plants	Thermal & nuclear power plants
Cooling	Natural or air cooling	Hydrogen or forced cooling (in turbo generators)

Source: EM_Module 3_Part1 (1).pdf

SYNCHRONOUS GENERATOR CONSTRUCTION

- salient 8-pole synchronous machine rotor



Source: EM_Module 3_Part1 (1).pdf

SYNCHRONOUS GENERATOR CONSTRUCTION

- **Rotor** experience varying magnetic fields, therefore is constructed of thin laminations to reduce eddy current losses
- To supply the rotor winding while it is rotating, special arrangement employed to connect its terminal to dc supply

1. supply dc power from an external dc source to rotor by means of slip rings
2. supply dc power from a special dc power source mounted on shaft of rotor

Source: EM_Module 3_Part1 (1).pdf

Fig. 37.6

paper.

When the conductor is in the interpolar gap, as at *A* in Fig. 37.7, it has minimum e.m.f. induced in it, because flux density is minimum there. Again, when it is at the centre of a *S*-pole, it has maximum e.m.f. induced in it, because flux density at *B* is maximum. But the direction of the e.m.f. when conductor is over a *N*-pole is opposite to that when it is over a *S*-pole.

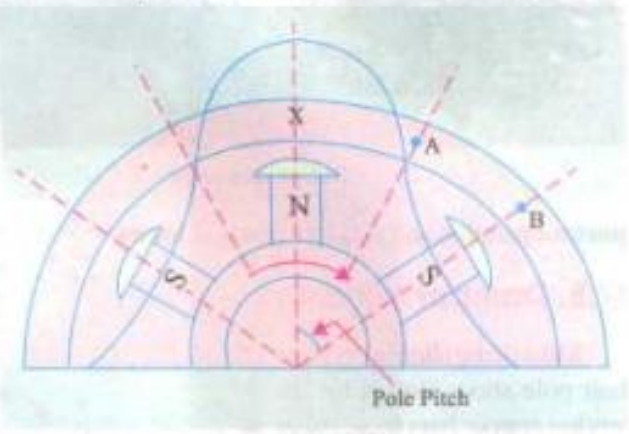


Fig. 37.7

Obviously, one cycle of e.m.f. is induced in a conductor when one pair of poles passes over it. In other words, the e.m.f. in an armature conductor goes through one cycle in angular distance equal to twice the pole-pitch, as shown in Fig. 37.7.

Let P = total number of magnetic poles

N = rotative speed of the rotor in r.p.m.

f = frequency of generated e.m.f. in Hz.

Since one cycle of e.m.f. is produced when a pair of poles passes past a conductor, the number of cycles of e.m.f. produced in one revolution of the rotor is equal to the number of pair of poles.

$\therefore$ No. of cycles/revolution = $P/2$ and No. of revolutions/second = $N/60$

$$\therefore \quad \text{frequency} = \frac{P}{2} \times \frac{N}{60} = \frac{PN}{120} \text{ Hz}$$

$$\text{or} \quad f = \frac{PN}{120} \text{ Hz}$$

N is known as the synchronous speed, because it is the speed at which an alternator must run, in

Source: EM_Module 3_Part1 (1).pdf

SYNCHRONOUS GENERATOR

Speed of rotation of synchronous generator

- synchronous generators are *synchronous*, *during their operation*

means: electrical frequency is synchronized with mechanical speed of rotor

- Relation between electrical frequency of stator and mechanical speed of rotor as shown before:
- $f = Ns\ p / 120$

f : electrical frequency in Hz

Ns: speed of rotor in r/min

p: number of poles

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Winding factor

The product of pitch factor K_p and distribution factor K_d is referred to the winding factor

$$i.e.\ K_w = K_p K_d.$$

The pitch factor and di...

Source: EM_Module 3_Part1 (1).pdf

Pitch factor

3. 6.1. Pitch Factor. In a short pitch winding the induced emfs in the two sides of the coil are not in phase, as discussed in Art 3.5, hence their resultant given by the phasor sum is always less than their arithmetic sum. The ratio of phasor sum of induced emfs per coil to the arithmetic sum of induced emfs per coil is known as *coil pitch factor, coil-span factor* or simply *pitch factor* (K_c or K_p). It is always less than unity.

Let the coil have a pitch short by an angle α electrical space degrees from full pitch and induced emf in each coil side be E.

Now if the coil would have been full-pitched, then total induced emf in the coil would have been 2 E.

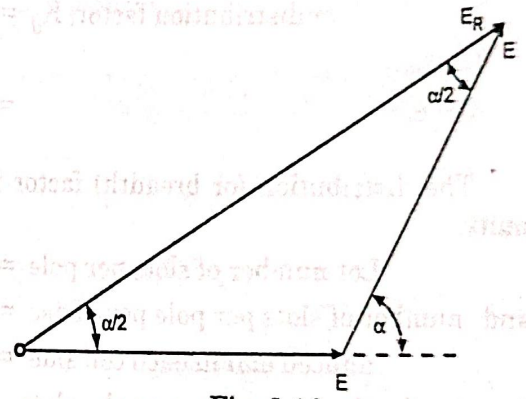


Fig. 3.16

If the coil is short-pitched by α electrical space degrees the resultant induced emf, E_R in the coil is phasor sum of two voltages α° (electrical) apart

$$i.e.\ E_R = 2\,E\cos\frac{\alpha}{2}$$

$$\text{Pitch factor, } K_p = \frac{\text{Phasor sum of coil side emfs}}{\text{Arithmetic sum of coil side emfs}} = \frac{2\,E\cos\frac{\alpha}{2}}{2E} = \cos\frac{\alpha}{2} \qquad ...(3.3)$$

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Distribution factor

The ratio of the phasor sum of the emfs induced in all the coils distributed in a number of slots under one pole to the arithmetic sum of the emfs induced (or to the resultant of emfs induced in all coils concentrated in one slot under one pole) is known as breadth factor (K_b) or distribution factor (K_d)

$$\text{or distribution factor, } K_d = \frac{\text{EMF induced in distributed winding}}{\text{EMF induced if the winding would have been concentrated}}$$

$$= \frac{\text{Phasor sum of component emfs}}{\text{Arithmetic sum of component emfs}}$$

The distribution (or breadth) factor is always less than unity.

Let number of slots per pole $= n$

and number of slots per pole per phase $= m$

Induced emf in each coil side $= E_c$

Angular displacement between the slots,

$$\beta = \frac{180^\circ}{n}$$

The emfs induced in different coils of one phase under one pole are represented by side AC, CD, DE, EF, ...which are equal in magnitude (say each equal to E) and differ in phase (say by β°) from each other.

If bisectors are drawn on AC, CD, DE, EF... they would meet at common point (say at O). The point O would be the circum centre of the circle having AC, CD, DE, EF...as the chords and representing the emfs induced in the coils in different slots.

$$\text{EMF induced in each coil side, } E = AC = 2\,OA \sin\frac{\beta}{2}$$

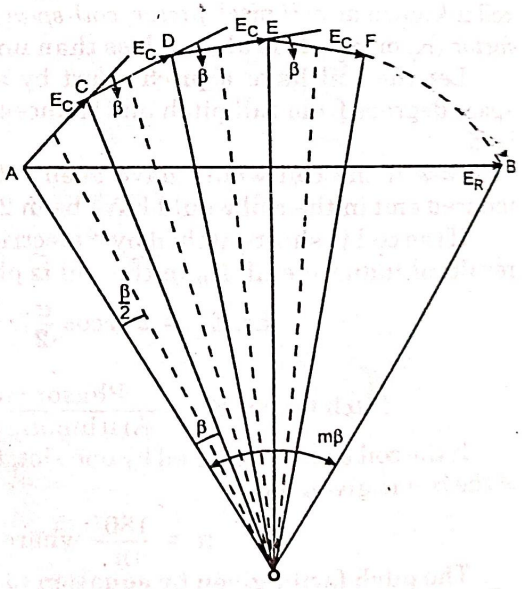


Fig. 3.17

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$$\text{Arithmetic sum} = m \times 2 \times OA \sin \frac{\beta}{2}$$

The resultant emf induced in one polar group of one phase would be the phasor sum as represented by the phasor AB shown in fig 3.17.

$$\therefore \text{The resultant emf, } E_R = AB = 2 \times OA \sin \frac{AOB}{2} = 2 \times OA \sin \frac{m\beta}{2}$$

$$\text{and distribution factor, } K_d = \frac{\text{Phasor sum of components emfs}}{\text{Arithmetic sum of components emfs}}$$

$$= \frac{2 \times OA \sin \frac{m\beta}{2}}{m \times 2 \times OA \sin \frac{\beta}{2}} = \frac{\sin \frac{m\beta}{2}}{m \sin \frac{\beta}{2}} \quad \ldots(3.5)$$

$m\beta$ is also known as the *phase spread* and is spread in electrical degrees or radians.

The distribution factor K_d given by equation (3.5) is for the fundamental component of emf. If the flux distribution contains space-harmonics, the slot angular pitch β on the fundamental scale, would become $r\beta$ for the *r th* harmonic component and thus the distribution factor for the *r th* harmonic would be

$$K_{dr} = \frac{\sin \frac{rm\beta}{2}}{m \sin \frac{r\beta}{2}} \quad \ldots(3.6)$$

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EMF Equation

Let Z_p = Number of conductors in series per phase
P = Number of poles
ϕ = Useful flux per pole in webers
N = Rotational speed in rpm

and f = Frequency in herz and is equal to $\frac{NP}{120}$

The flux cut by any conductor while passing from the centre of one inter polar gap to the centre of the next is ϕ webers and since during the movement, the emf wave completes half cycle *i.e.* the time taken is $\frac{1}{2f}$ seconds, therefore, the average rate of cutting the flux,

$$\frac{d\phi}{dt} = \frac{\phi}{1/2f} = 2f\phi \text{ Wb/s}$$

Hence average emf induced in each conductor = $2f\phi$ volts

Average emf per phase, E_{av}/phase = Number of conductors in series/phase arranged in one slot/pole × average emf induced per conductor

$= Z_p \times 2f\phi = 2T \times 2f\phi = 4\phi f T$ volts $\quad \because Z_p = 2T$ (turns)

For distributed winding the average value of emf per phase will be K_d times the above value

i.e. E_{av} / phase $= 4K_d \phi f T$ volts

For short pitch winding the true average value of emf per phase will be K_p time above value

E_{av}/phase $= 4K_d K_p \phi f T$ volts

and E_{rms}/phase = Form factor, $K_f \times E_{av}$/phase $= 4K_f K_d K_p \phi f T$ volts

For sinusoidal wave of emf, $K_f = 1.11$

$\therefore E_{rms}$ / phase $= 4.44 K_d K_p \phi f T$ volts ...(3.7)

Part 1 • Slide 35

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If the alternator is star-connected, as is usually the case, the line voltage is $\sqrt{3}$ times the phase value.

So line induced emf, $E_L = \sqrt{3} \times 4.44 K_d K_p \phi f T$ volts ...(3.8)

For full-pitched and concentrated windings, $K_p = 1$ and $K_d = 1$

Part 2: Original Slides

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PCEOT402
ELECTRICAL MACHINES

Module 3

Synchronous Machines

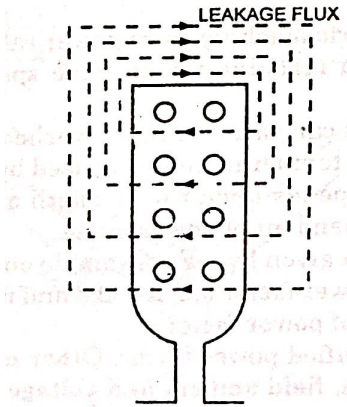
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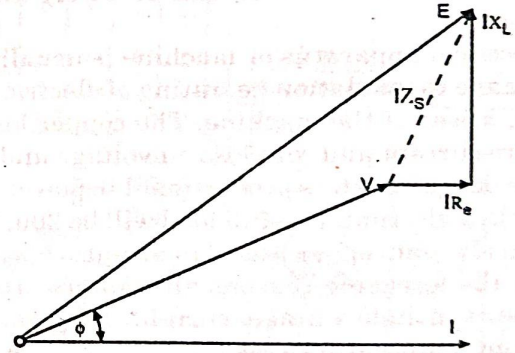
Three phase Synchronous Machine

- Synchronous generator on no-load – open circuit characteristics –
- synchronous generator on load – armature reaction – effect of armature reaction
- Equivalent circuit - phasor diagram – voltage regulation –
- predetermination of voltage regulation by emf and mmf method
- Parallel operation - synchronous generator on infinite bus-bar –
- conditions – methods of synchronisation – dark lamp method – bright lamp method

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Leakage reactance





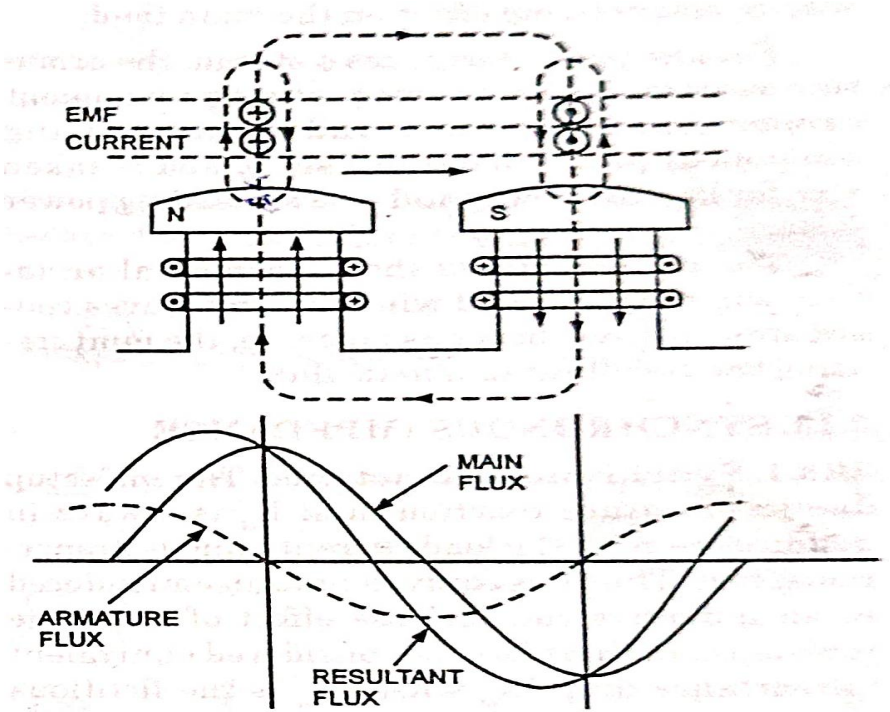
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Armature Reaction

- Effect of armature flux on main flux
- In alternator it depend on type of load
- Effects are
  - Magnetising
  - Demagnetising
  - Cross magnetising

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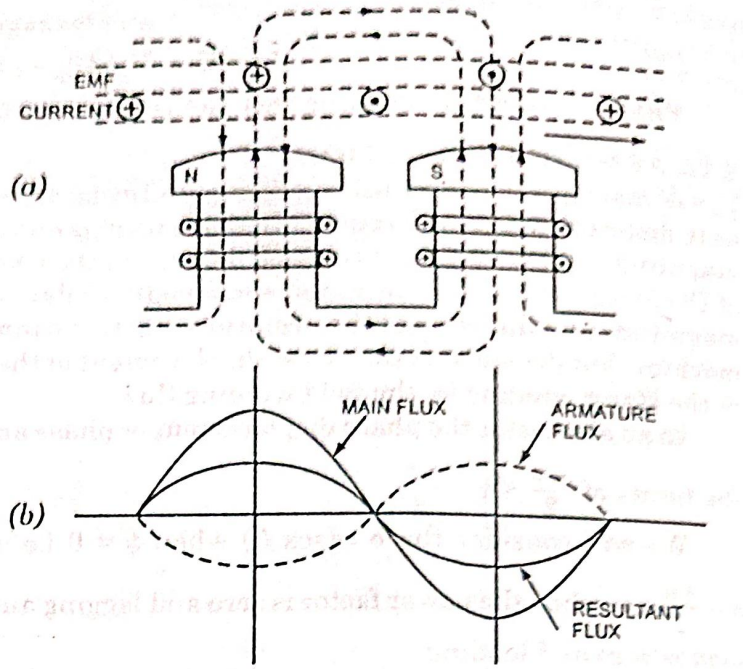
i) At upf load



Hence armature reaction at unity power factor has got distorting effect.

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ii) Zero power factor lagging

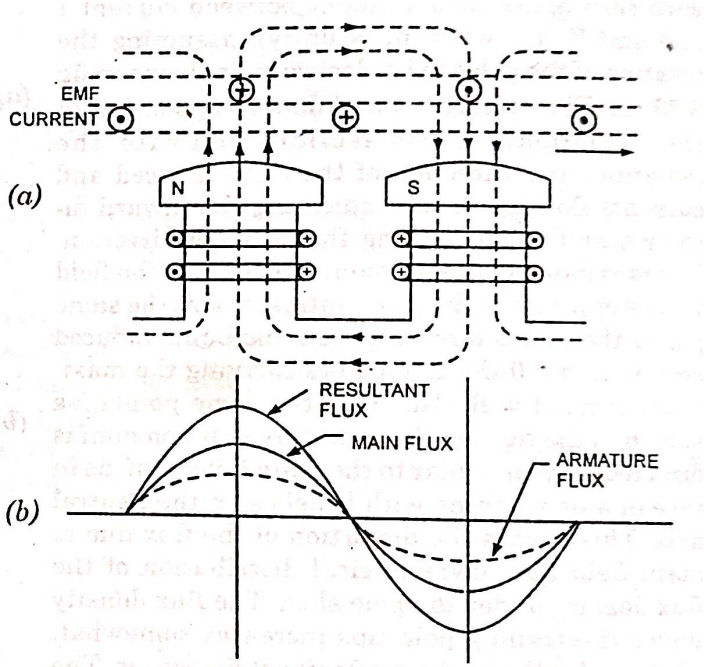


Armature Reaction At Zero Power Factor (Lagging)

demagnetising effect.

Source: EM_Module 3_Part2.pdf

iii) Zero power factor leading



Armature Reaction At Zero Power Factor (Leading)

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Synchronous Impedance

- Synchronous Reactance

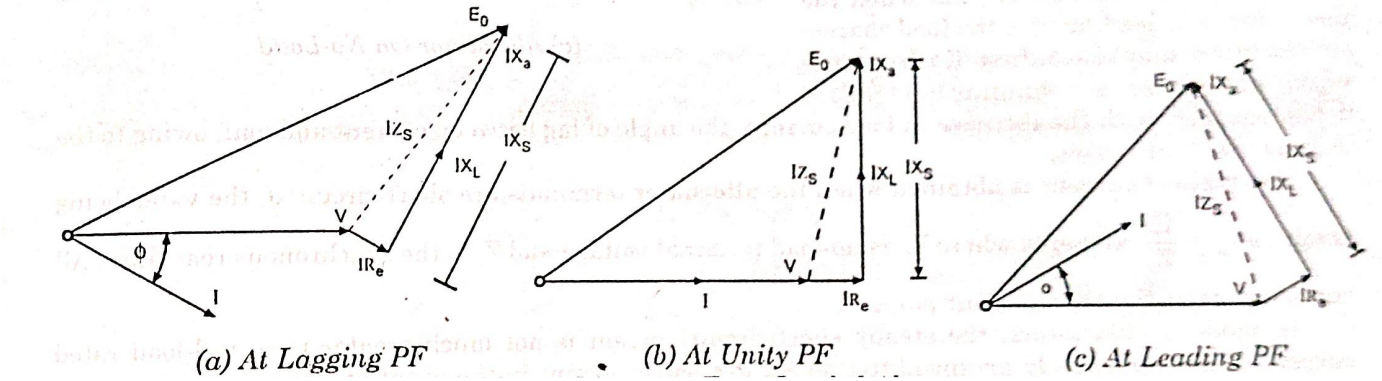
$$X_s = X_L' + X_a$$

- Synchronous Impedance

$$Z_s = R_a + jX_s$$

Source: EM_Module 3_Part2.pdf

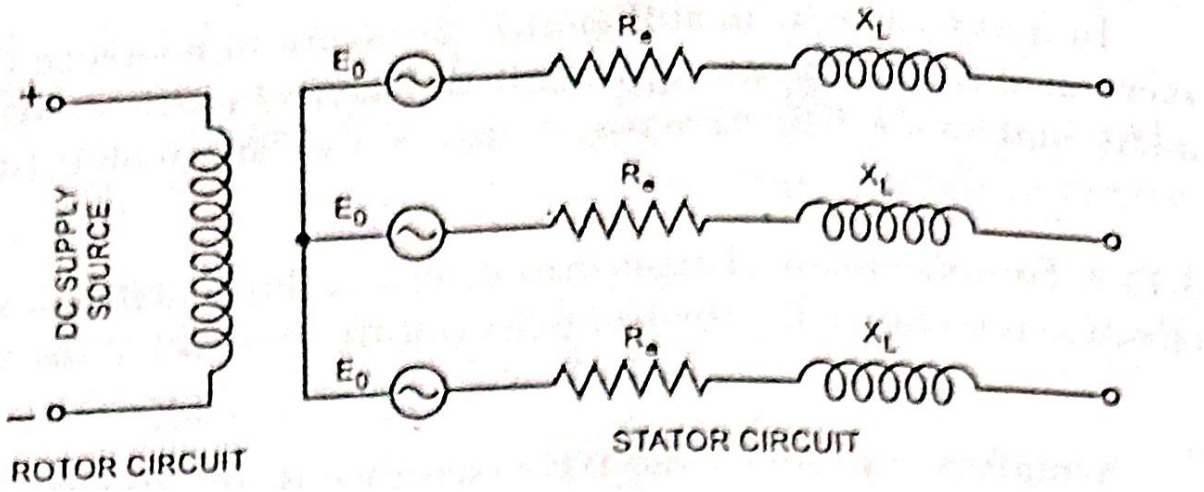
Alternator on load Phasor Diagram



(a) At Lagging PF *(b) At Unity PF* *(c) At Leading PF*

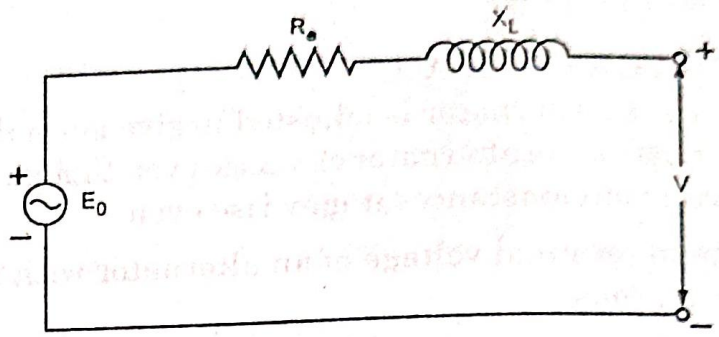
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Equivalent Circuit

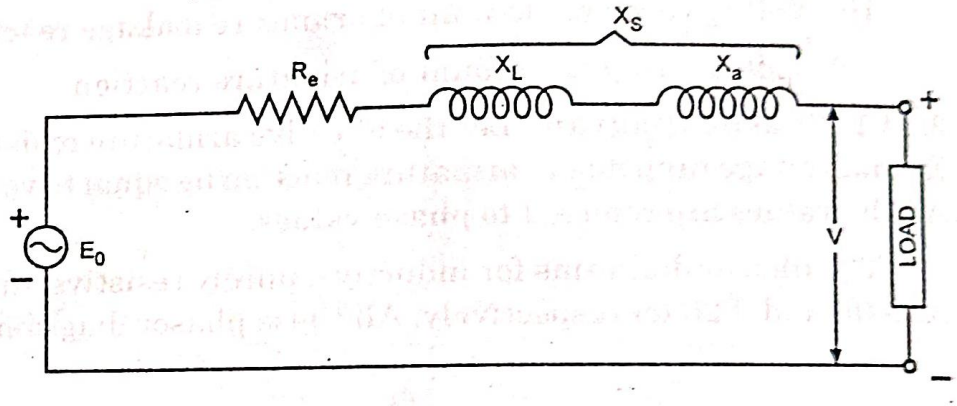


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Source: EM_Module 3_Part2.pdf



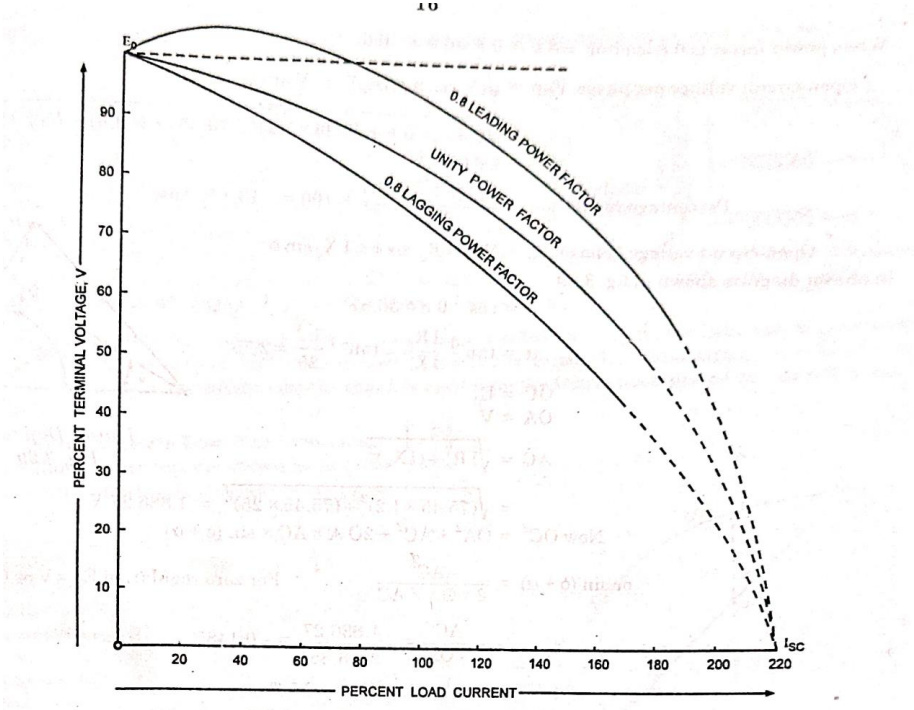
(b) Equivalent Circuit Diagram



(c) Alternator On No-Load

Source: EM_Module 3_Part2.pdf

Load Characteristics



Source: EM_Module 3_Part2.pdf

Voltage Regulation

$$\textit{i.e.}\ \text{voltage regulation} = \frac{E_0 - V}{V}\ \text{per unit}$$

$$= \frac{E_0 - V}{V} \times 100\ \text{in percentage}$$

Source: EM_Module 3_Part2.pdf

EMF Method (Synchronous Impedance Method)

nd short-circuit characteristic.

From short-circuit and open-circuit tests, the voltage egulation can be determined by any one of the two methods iven below.

1. Synchronous Impedance Method or EMF lethod. After determining R_e and X_s from open-circuit and ort-circuit tests, the phasor diagram for any load of any wer factor can be drawn, as shown in fig 3.35.

Let load current be I lagging behind the terminal phase ltage V by ϕ

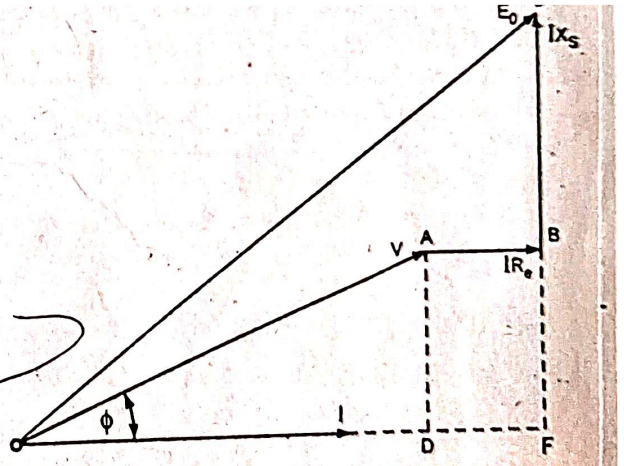


Fig. 3.35

No load terminal voltage per phase, E_0 = OC

Fulll-load terminal voltage, V = OA

Armature resistance drop, $I R_e$ = AB

Synchronous reactance drop $I X_s$ = BC

$$\text{Now } OC^2 = OF^2 + FC^2 = (OD + DF)^2 + (FB + BC)^2$$

$$\text{or } E_0^2 = (V \cos \phi + I R_e)^2 + (V \sin \phi + I X_s)^2$$

[Since BF = AD = V sin ϕ]

Source: EM_Module 3_Part2.pdf

$$\text{or } E_0 = \sqrt{(V\cos\phi + IR_e)^2 + (V\sin\phi + IX_s)^2}$$

$$\text{And \% regulation} = \frac{E_0 - V}{V} \times 100$$

Note 1. For leading pf taking ϕ as negative and for unity pf taking $\phi = 0$ voltage regulation determined at leading pf and unity pf respectively.

2. Of various methods for determination of regulation, this one yields results most likely to be The regulation obtained by this method is always higher than actual value and, therefore, this is ca *pessimistic* method. The results are, however, more likely to be 'on safe side'. However, this m theoretically accurate for non-salient pole machines with distributed field windings when saturati considered. The reason of error is that in this method synchronous impedance or reactance is ass remain constant while actually it is not. The value of synchronous impedance (or reactance) varies saturation. At low saturation its value is larger because the effect of armature reaction is greater th at high saturation. Now under short-circuit conditions, saturation is very low and, therefore, the synchronous impedance measured is higher than that in actual working conditions.

Source: EM_Module 3_Part2.pdf

MMF (Ampere Turn Method)

synchronous impedance measured is higher than that in actual working conditions.

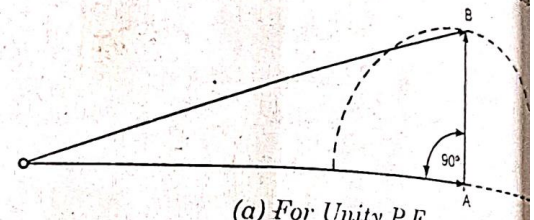
2. *Ampere-Turn (or MMF) Method.* In this method, the datas determined from open-circuit te utilized.

From the above two tests the open-circuit and short-circuit characteristics are drawn, as shown in fig 3.32.

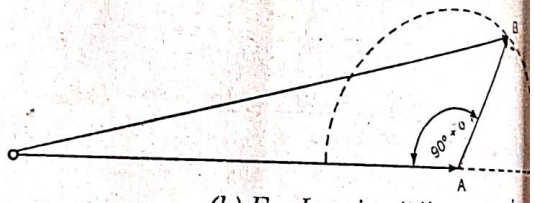
From the above characteristic field current I_{f1} is determined to give rated voltage V on no load, neglecting armature resistance drop, and field current I_{f2} is determined to cause short-circuit current, equal to full load current, on short circuit.

On short circuit, the field excitation I_{f2} balances the impedance drop in addition to armature reaction on full load. But since R_e is usually very small and X_L is also small for low voltage on short circuit, so impedance drop can be neglected. Hence pf on short circuit is almost zero lagging and the field amp-turns are used entirely to overcome the armature reaction. Therefore, I_{f2} gives demagnetising amp-turns at full load.

Now let the alternator supply full load current at a pf of cos ϕ. Draw OA representing I_{f1} to give full load rated voltage, V [or more exactly V + $I R_e \cos \phi$] then draw AB at an angle $(90^\circ \pm \phi)$ representing I_{f2} to give full load current on short circuit; + ve sign for lagging pf and –ve sign for leading pf. Now find field current I_f measuring OB, which will give open-circuit EMF E_0, which can be determined from OCC



(a) For Unity P F



(b) For Lagging P F



(c) For Leading P F

Source: EM_Module 3_Part2.pdf

Parallel Operation of Alternators

- Efficiency
- Reliability or Continuity of Service
- Maintenance and Repair
- Physical Size

Source: EM_Module 3_Part2.pdf

Requirements for Parallel Operation

- ✓ Alternators must have same output voltage rating
- ✓ Speeds of the machines should be such as to generate same frequency
- ✓ Alternators should be of same type so as to generate voltages of same waveform
- ✓ Prime movers of the alternators should have same speed-load characteristics
- ✓ Alternators should have reactances in their armature

Source: EM_Module 3_Part2.pdf

Conditions for Synchronising

* **Synchronising** - The process of connecting an alternator in parallel with another alternator or with the common bus bar.

* For satisfactory operation,

- The terminal voltage of incoming machine must be exactly equal to that of running machines or common bus bar
- The speed of incoming machine must be such that its frequency *f*

equals to bus bar frequency

- Phase of incoming machine voltage must be the same as that of the bus bar voltage relative to the load.
- For three phase alternator, the phase sequence of incoming machine must be same as that of bus bar.

Source: EM_Module 3_Part2.pdf

Methods of Synchronisation

- Synchronising of three phase alternators
  - Three Dark Lamp Method
  - Bright lamp Method (Two Bright and One Dark Lamp Method)
  - Synchroscope

Source: EM_Module 3_Part2.pdf

Synchronization of 1-Phase Alternators

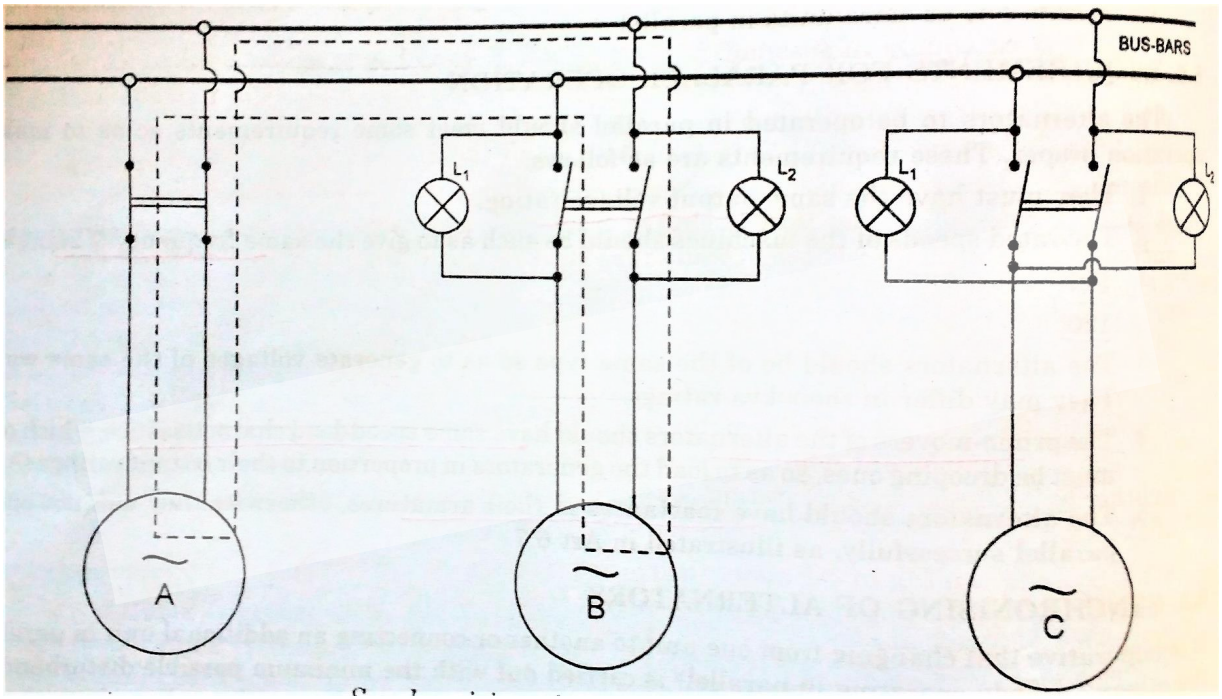


Figure 8 : Synchronising of single phase alternators

Source: EM_Module 3_Part2.pdf

Dark Lamp Method

- Equality of terminal voltages can be determined by connecting voltmeters to the incoming and running alternators.
- Equality of frequency and phasing can be determined with the help of synchronising lamps.

Let alternator B is to be paralleled with alternator A,

- Prime mover of Alternator B is started and brought up close to the rated speed.
- Alternator B is excited and its voltage is raised by increasing the excitation.
- If frequencies of alternators A & B are exactly same and their terminal voltages w.r.t local series circuit → no resultant voltage will act across lamps L_1 and L_2 → Lamps will remain dark.
- If frequencies of A & B are not equal → flickering of lamps. Frequency of flickering = Difference of frequencies of two alternators.
- $\Rightarrow$ greater the difference in frequencies, greater will be the frequency of flick

Source: EM_Module 3_Part2.pdf

Dark Lamp Method

- Synchronising is done at the middle of dark period.

  "At the time of synchronising of alternators, the speed of the incoming machine is adjusted until the lamps go in and out very slowly, the terminal voltage of the incoming alternator is made equal to bus bar voltage by adjusting the excitation of the incoming alternator and then switch of incoming alternator is closed in the middle of the dark period."

- But it is not so easy to judge the middle of dark period.

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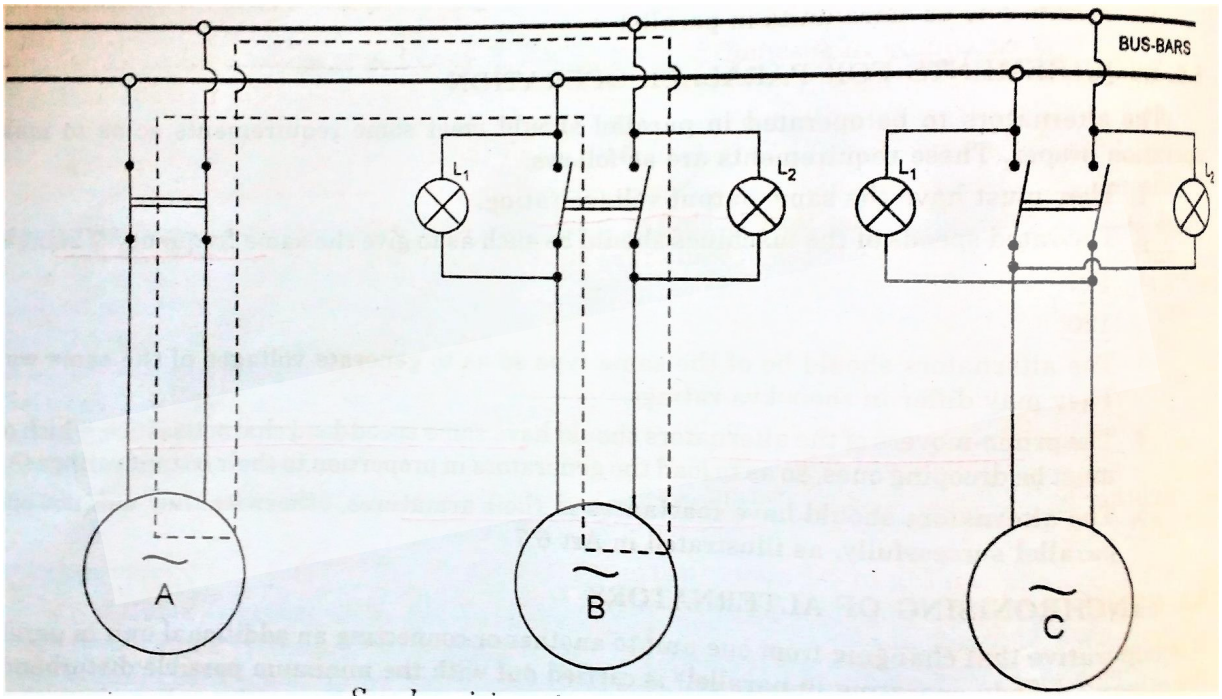


Figure 9 : Synchronising of single phase alternators

Source: EM_Module 3_Part2.pdf

Bright Lamp Method

- Easy to judge the middle of bright period
- In **Bright lamp method**, lamps are cross-connected.
- Maximum voltage across lamps will occur when alternator C is in phase opposition with alternator A.
- Magnitude of maximum voltage that can exist across a lamp is double that of the voltage of one machine.
- After doing the adjustments, the synchronising switch is closed at the middle of bright period.

Source: EM_Module 3_Part2.pdf

Synchronization of 3-Phase Alternators

"Phasing out the alternator" → The process of checking the phase sequence and getting it correct

1. Three Dark Lamp Method
2. Two Bright and One Dark Lamp Method
3. Using Synchroscope

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Source: EM_Module 3_Part2.pdf

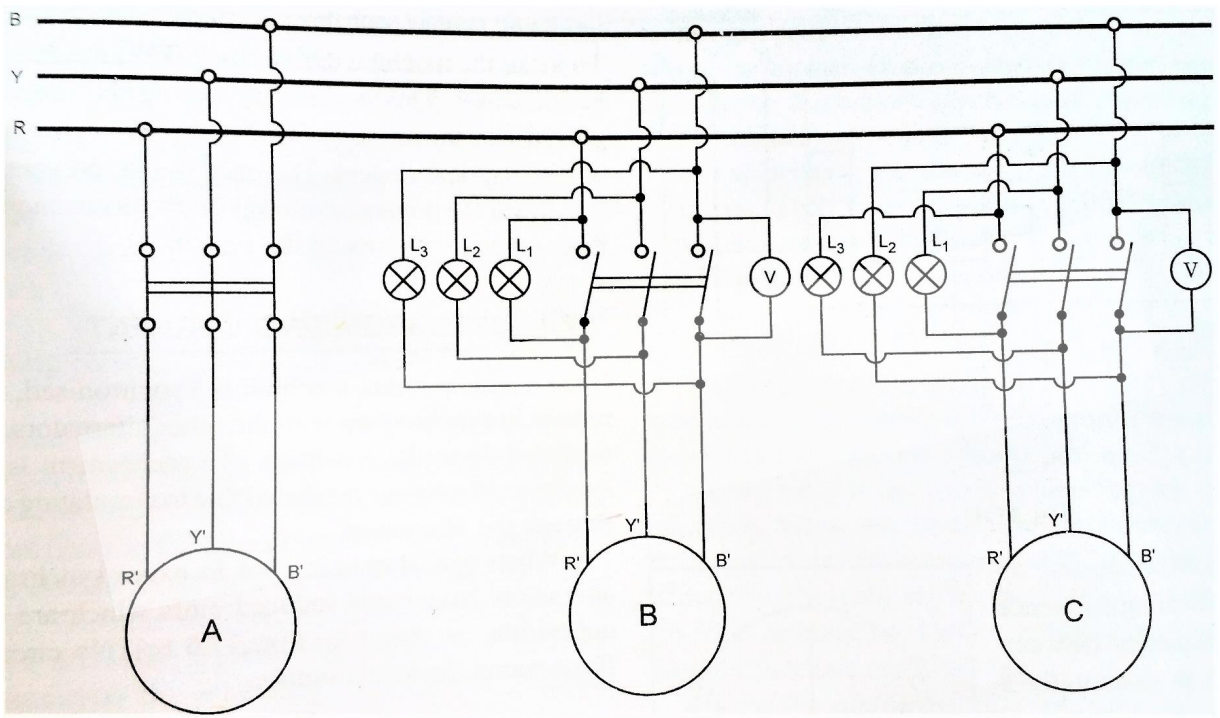


Figure 10 : Synchronising of 3-Phase Alternators

Source: EM_Module 3_Part2.pdf

Three Dark Lamp Method

- Prime mover of alternator B is started and brought up close to the rated speed.
- Then alternator B is excited and its voltage is raised by increasing the excitation.
- If the incoming alternator B is properly connected, all the three lamps should become bright and dark together.
- If they bright and dim in sequence ⇒ incoming alternator B is not properly connected with the bus-bars and the phase sequence of incoming alternator B must be reversed relative to the system.(Interchange any two leads on either the alternator side or the line side of the switch)
- The speed of incoming machine is further adjusted until the lamps flicker at a very rate and voltage is made equal to the bus bar voltage by adjusting the excitation.
- The synchronising switch is closed at the instant all the three lamps

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Source: EM_Module 3_Part2.pdf

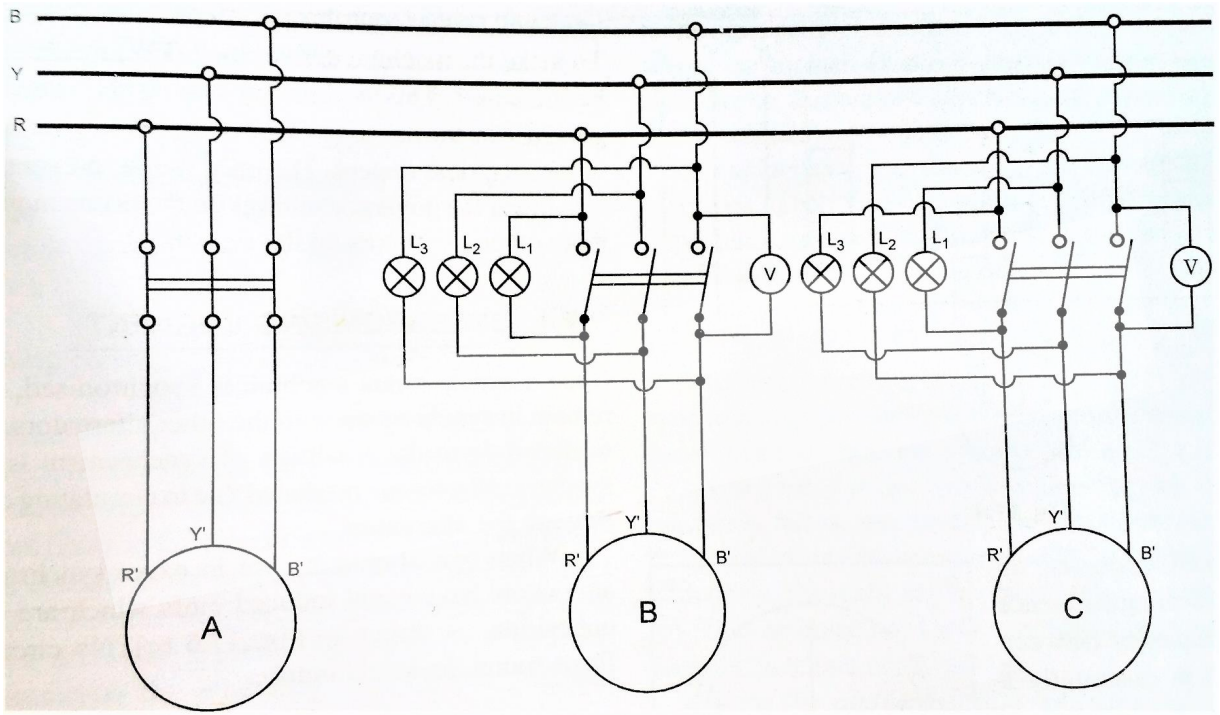


Figure 11 : Synchronising of 3-Phase Alternators

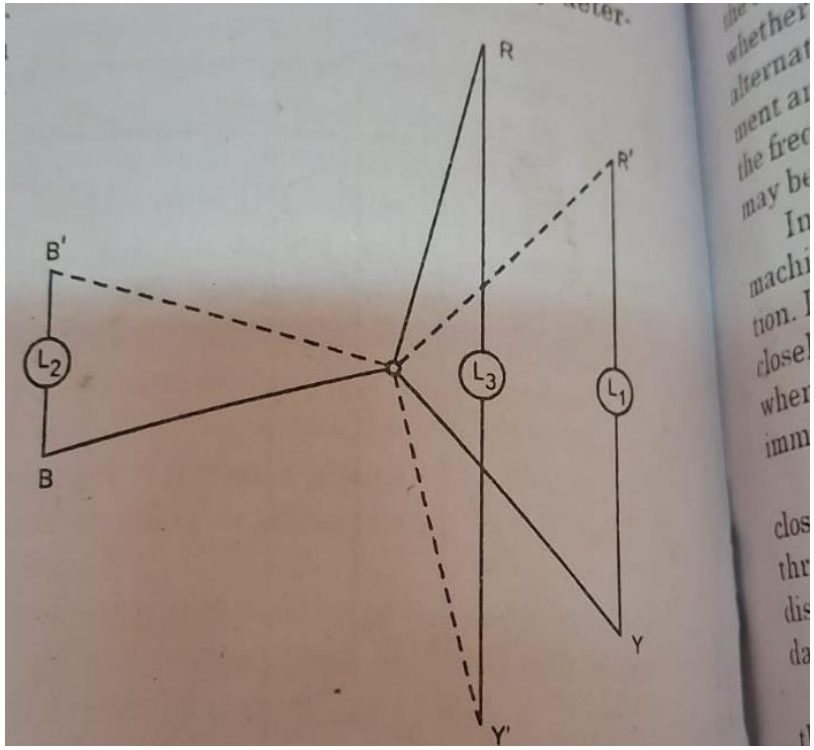
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Two Bright and One Dark Lamp Meth

- When incoming alternator is in synchronism with running machine, lamps L_1 and L_3 are bright and L_2 is dark.
- Since near the point of synchronism, the brightness of one lamp increases and of another decreases. The instant at which the incoming machine is in synchronism with the bus-bars can be accurately determined and the paralleling switch will be closed at this instant.
- If the incoming machine is too fast → voltage across L_3 decreases, L_1 increases and L_2 decreases.
- If the incoming machine is too slow → voltage across L_3 increases, L_1 decreases and L_2 increases.
- Hence when the three lamps are placed in a ring, a light wave travelling in counter-clockwise direction indicates that the incoming machine is slow and light wave travelling in clockwise direction indicates that the incoming machine is fast.
- Synchronising switch is closed when changes in light are very slow and at the instant the lamp L_2 is dark.

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Source: EM_Module 3_Part2.pdf



Source: EM_Module 3_Part2.pdf

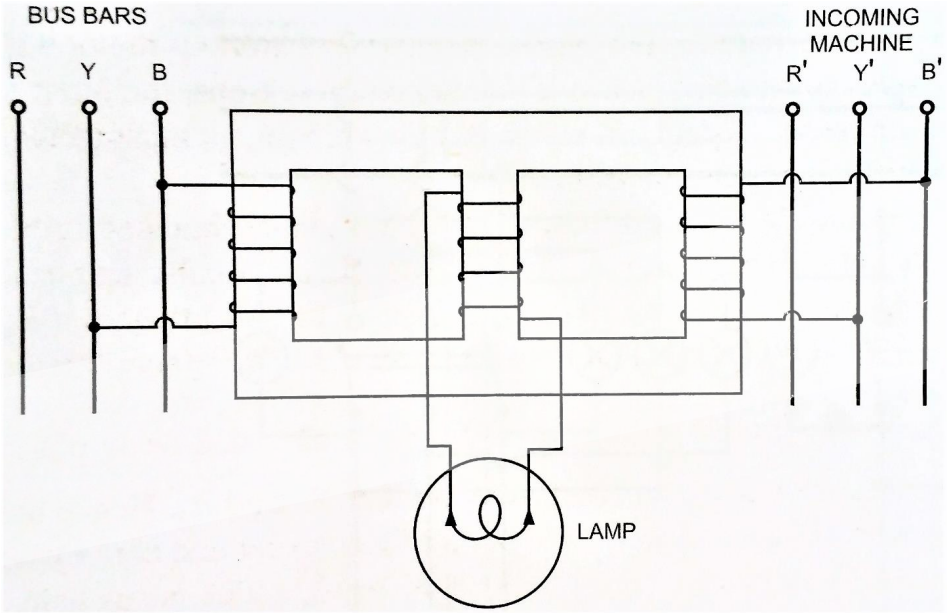


Figure 12 : Three Limbed Transformer and Lamp

Source: EM_Module 3_Part2.pdf

Synchroscope

- Synchroscope → Instrument indicating the difference of phase and frequency between two voltages.
- Its a split-phase motor
- Torque is developed if the frequencies of the two voltages differ.
- Voltages from corresponding phases of incoming and running alternators are applie to synchroscope.
- When frequencies are equal, no torque is exerted on the pointer and the pointer stops at the middle of the dial.
- When pointerstops at vertical position, the frequencies are equal, the voltages are in phase and the paralleling switch can be closed.

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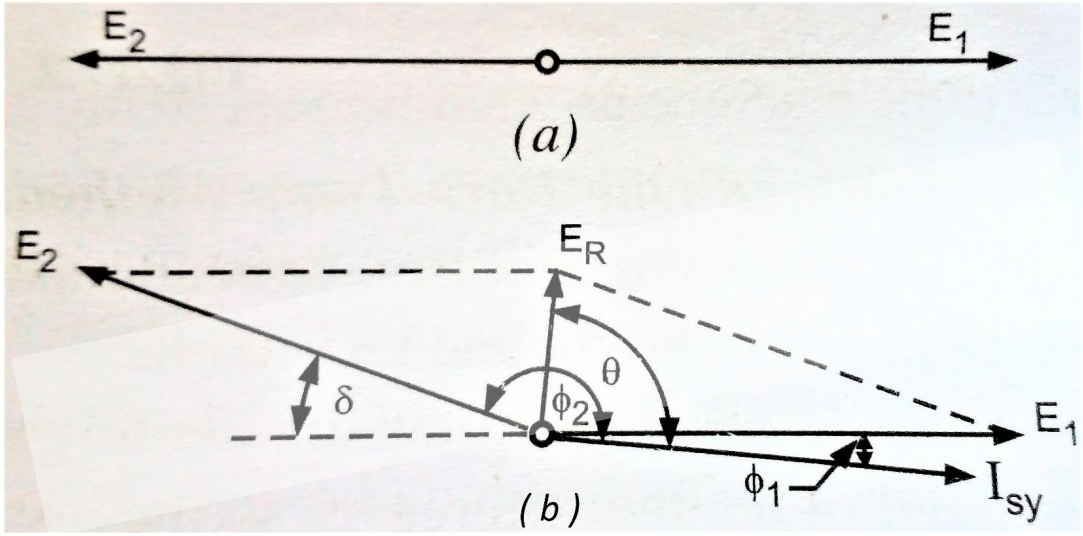


Figure 15 : Voltage Phasors

Source: EM_Module 3_Part2.pdf

Synchronising Current

- When two alternators are in synchronism → have equal emfs in exact phase opposition → no circulating current (Fig. (a))
- When induced emfs are equal in magnitude but not in exact phase opposition → resultant emf around the local circuit /= zero → causes flow of current → **Synchronising current(I_{sy})**

  → Let alternator-2 tends to retard,
- E_2 falls back by a phase angle δ electrical degrees
- $|E_1| = |E_2| = E$
  but phase difference = 180 - δ

Source: EM_Module 3_Part2.pdf

→ Resultant emf(E_R)

$$E_R = 2E\cos\frac{180 - \delta}{2}$$

$$= 2E\cos\left(90 - \frac{\delta}{2}\right)$$

$$= 2E\sin\frac{\delta}{2}$$

$$= 2E \times \frac{\delta}{2}$$

$$= E\delta$$

→ Synchronising current(I_{sy})

$$\frac{E_R}{Z_{cs}} \quad \frac{E\delta}{Z_{cs}}$$

Source: EM_Module 3_Part2.pdf

Synchronising Current

- I_{sy} lags behind E_R by an angle θ
- If $R_{ce} <<< X_{cs}$,

$$I_{sy} = \frac{E_R}{X_{cs}}$$

$\Rightarrow$ I_{sy} lags E_R by 90° $\Rightarrow$ almost in phase with E_1

- I_{sy} → generating current w.r.t machine 1 and motoring current w.r.t machine 2
- I_{sy} sets up synchronising torque(T_{sy}) which tends to accelerate machine no.2 and decelerate machine no.1
- Any departure from synchronism results in development of synchronising torque which tends to keep the machines in synchronism

Source: EM_Module 3_Part2.pdf

Synchronising Power

- Power supplied by Machine no.1 = $E_1 I_{sy} Cos\Phi_1$
- Power received by Machine no.2 = $E_2 I_{sy} Cos\ (180 - \Phi_2)$
- Power supplied by Machine no.1 = Power received by Machine no.2 + copper losses
- Power supplied by Machine no.1 is called as **Synchronising power(P_{sy})**

$$P_{sy} = E_1 I_{sy} Cos\Phi_1$$

$$= E_1 I_{sy} \text{ if } \delta$$

$$= E \times \frac{E\delta}{X_{cs}}$$

$$= \frac{\delta E^2}{X_{cs}}$$

- Total synchronising power = $3P_{sy} = \frac{3\delta E^2}{X_{cs}}$

Source: EM_Module 3_Part2.pdf

Synchronising Torque

$$3P_{sy} = T_{sy} \times \frac{2\pi N_s}{60}$$

Therefore Synchronising torque,

$$T_{sy} = \frac{3P_{sy} \times 60}{2\pi N_s}$$

where N_S is the synchronous speed in rpm

Source: EM_Module 3_Part2.pdf

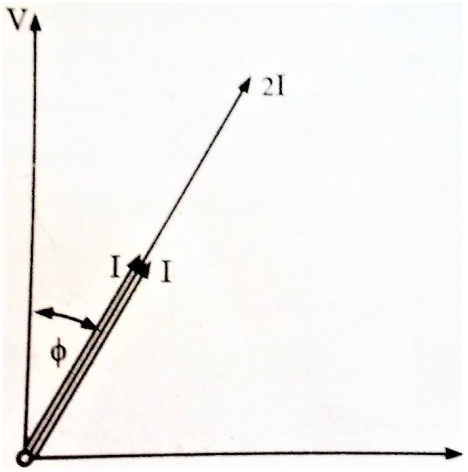


Figure 16 : (a)

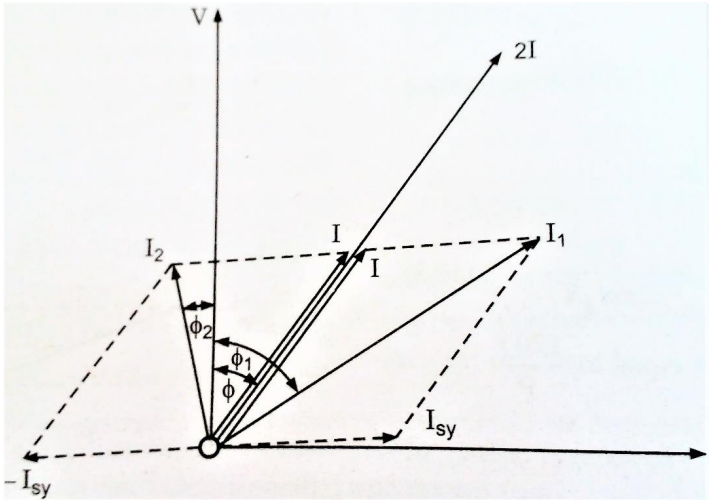


Figure 17 : (b)

Part 3: Original Slides

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MODULE III

Synchronous motor – Construction, Principle of operation, Effect of excitation on armature

Source: EM_Module 3_Part3.pdf

Synchronous Motor - Features

- By construction it is very similar to synchronous generator
- It is a constant speed motor – speed decided by frequency
- Its power factor is controllable
- It requires both ac and dc supplies
  - AC for armature
  - DC for field excitation
- Synchronous motors are not self starting type
- Used for applications were constant speed operation is essential
  - Example: steel rolling mills, paper rolling mills

Source: EM_Module 3_Part3.pdf

Construction of Synchronous motor

- It has mainly 2 parts
-

1. Stator
2. Rotor

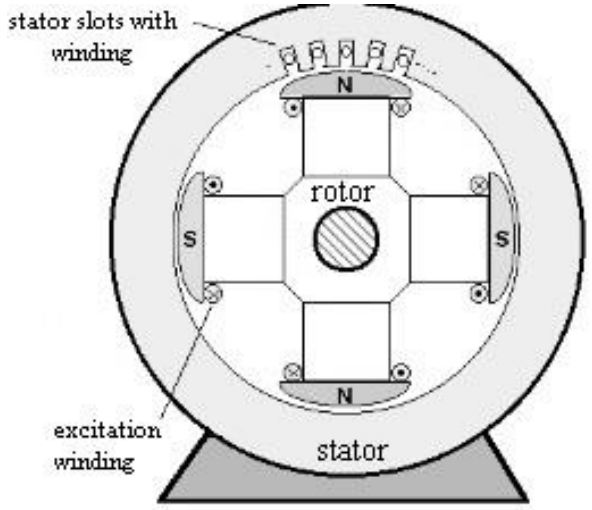
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1. Stator or Armature

- Stator is the stationary part
- Main parts of **stator** are ***stator core, stator winding & stator frame***
- ***Stator core*** contain an iron ring made of silicon steel laminations
- Slots are cut on the inner periphery of the stator core, in which stator conductors are placed
- Stator core is laminated to reduce eddy current loss
- axial & radial ventilating ducts to provide efficient cooling
- Slots can be open, semi closed or totally closed
- Open slots are commonly used

Source: EM_Module 3_Part3.pdf

- Stator contain **a 3 phase winding** which is normally star connected
- Stator conductors are made of copper or aluminium
- Windings can be single layer or double layer, lap or wave, concentrated or distributed
- The whole structure is held in **a frame** made of cast iron or welded steel plate



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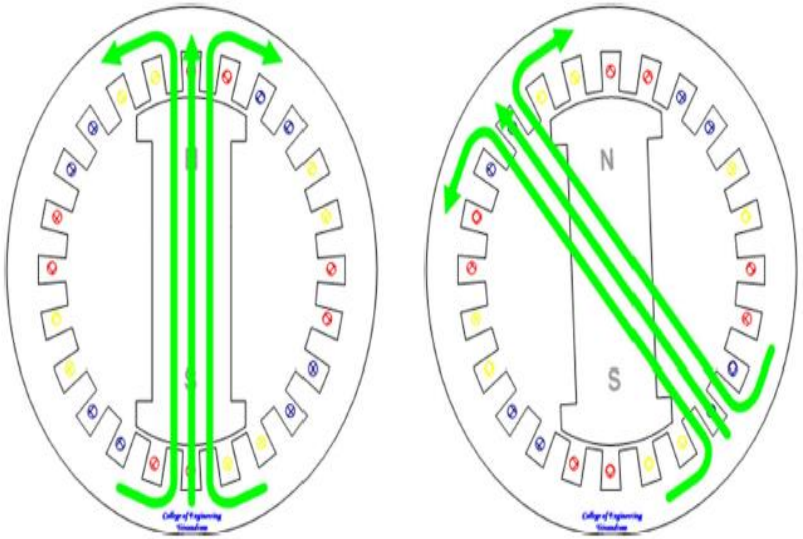
2. Rotor or Field magnet system

- It is the rotating part of the machine
- Rotor contain Field magnets & are supplied with DC voltage
- For this external DC sources are used & exciting current is supplied to rotor through slip rings & brushes
- An exciter mounted on the motor shaft can also be used to provide necessary excitation.
- Normally, the rotor is having a salient pole structure
- Field poles contains laminated pole core & pole shoe (to reduce eddy current losses) and field windings

Part 3 • Slide 7

Source: EM_Module 3_Part3.pdf

Why Synchronous Motors are not Self Starting ?



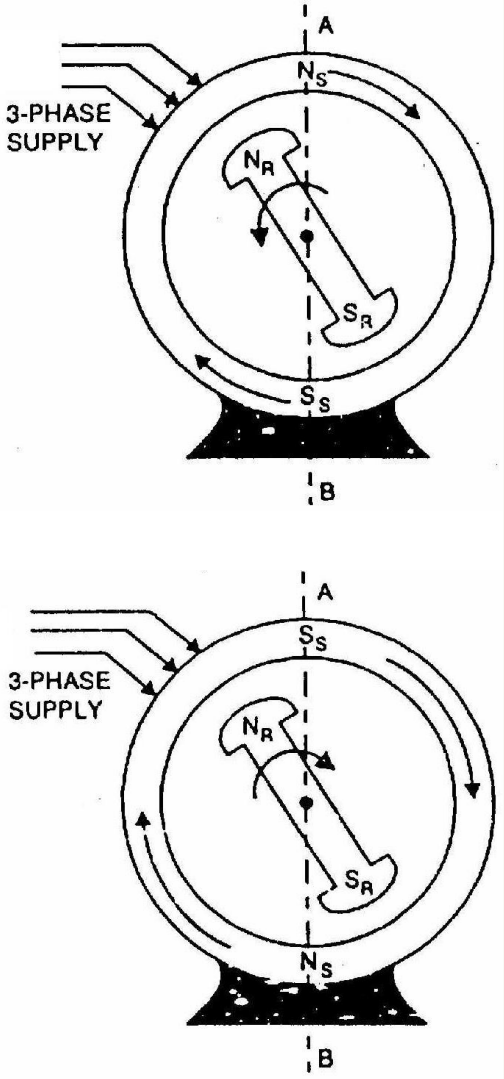
Due to the inertia of the rotor, it may not move quickly with the developed RMF when supply voltage is applied.

As the N and S poles of the RMF pass by, the rotor will tend to move in both clockwise and anticlockwise directions, but will not attain speed.

So synchronous motors are not self starting.

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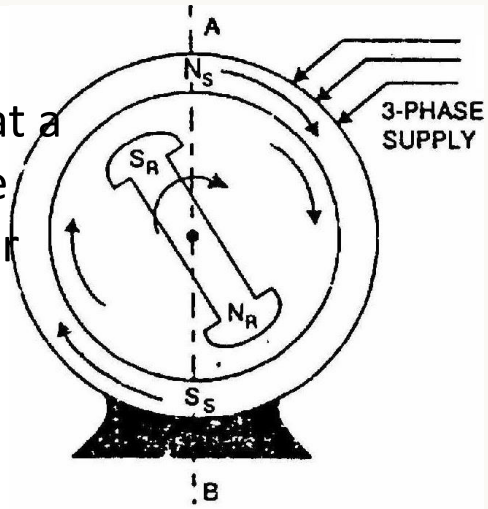
- Now stator North pole (Ns) & rotor North pole (Nr) will repel each other & rotor experiences a torque in anticlockwise direction
- After a period of half cycle, stator poles changes their position but rotor poles remain at previous position
- Now stator South pole (Ss) & rotor North pole (Nr) will attract each other & rotor experiences a torque in clockwise direction
- Since stator poles changes their polarity rapidly, rotor experiences torque in opposite directions in a cycle.
- Due to high inertia of rotor, motor fails to start

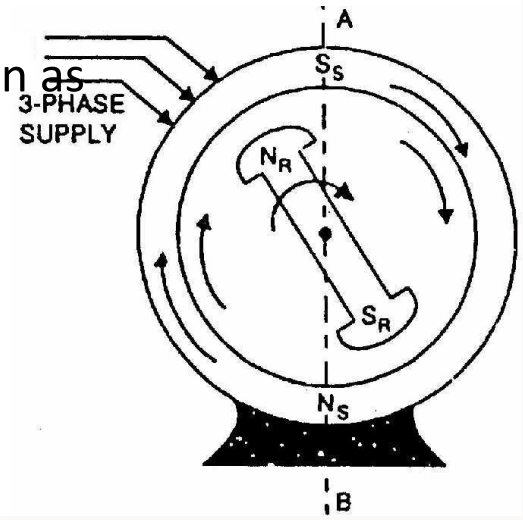


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How to get unidirectional torque?

- If rotor is rotated by some external means at a speed near synchronous speed, in the same direction of stator magnetic field, then rotor will experience a unidirectional torque
- Now stator South pole (Ss) & rotor North pole (Nr) will attract each other & rotor experiences a torque in clockwise direction
- After a period of half cycle, stator & rotor magnetic field poles changes their position as shown in second figure
- rotor experiences a torque in clockwise direction.





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Source: EM_Module 3_Part3.pdf

- Under this condition, stator & rotor poles of opposite polarity experiences a strong attraction & the mutual attraction locks the rotor & stator together
- Now if the external prime mover is removed, rotor will continue to rotate at synchronous speed due to magnetic locking between stator & rotor

Source: EM_Module 3_Part3.pdf

Starting methods of Synchronous motor

- Synchronous motor has no self starting torque & motor must first be brought upto synchronous speed by some external means for getting unidirectional torque in rotor.
- Various methods of starting of synchronous motors are

1. From DC source

- If a DC supply & a DC motor (shunt/compound) is available, the Synchronous motor is coupled to DC motor & is started using DC motor
- DC motor is coupled to Synchronous motor and dc motor is started & rotated at synchronous speed
- AC supply is given to Synchronous motor (make sure that rotating magnetic field & rotor of synchronous motor are rotating in same direction)
- Now DC field excitation is given & rotor gets locked to stator rotating magnetic field & continue to rotate at synchronous speed
- The DC motor is now disconnected from supply

Source: EM_Module 3_Part3.pdf

2. By using an AC motor

- Here a small AC induction motor (called pony motor) is used for starting the Synchronous motor
- First the induction motor is started & will rotate Synchronous motor rotor at near synchronous speed
- AC supply is given to Synchronous motor (make sure that rotating magnetic field & rotor of synchronous motor are rotating in same direction)
- Now DC field excitation is given & rotor gets locked to stator rotating magnetic field & continue to rotate at synchronous speed
- The Induction motor is now disconnected from supply

3. By providing damper windings

- Synchronous motor can be made self starting by providing a special winding on rotor poles known as damper winding or squirrel cage winding
- Damper winding consist of short circuited copper bars placed in rotor poles
- When AC supply is given to stator, the synchronous motor will start as an Induction motor & rotates at a speed near synchronous speed
- Now DC field excitation is given & rotor gets locked to stator rotating magnetic field & continue to rotate at synchronous speed

Source: EM_Module 3_Part3.pdf

Equivalent circuit of a Synchronous motor

- In a synchronous motor we are giving two supply, DC supply to rotor & 3 phase AC supply to stator
- When AC supply is given to stator, a current flows through stator windings
- Stator windings have resistance (R_a) & synchronous reactance (X_s)
- Also when a DC supply is given to rotor, a magnetic field is produced & it links stator conductors,
- As rotor rotates, flux linked with stator conductors changes & an EMF induces in stator windings. This is called back EMF (E_b) or counter EMF
- The back EMF opposes the stator supply voltage